Endpoint crossings for random walks with heavy-tailed runs and for aging walks

Arjun Pemmasani
Department of Mathematics, Harvey Mudd College
Claremont, CA 91711, USA
apemmasani@hmc.edu
ORCID 0009-0001-2776-6298

Abstract

Consider a random walk on $\mathbb{Z}$ whose direction is an alternating renewal process started in equilibrium, with independent run lengths of tail $k^{-\alpha}$, $1 < \alpha < 2$. This is a lattice analogue of a Lévy walk. We compare the probability of each end, $\pm N$, after N steps with that of the center. Wang, Li and Hänggi (2016) observed, as a heuristic for heat conduction, that the ends and the center decay at the same rate when α is the golden ratio. We prove the lattice form of their observation, with its constant. Namely, the ends beat the center for large N exactly when α is below the golden ratio, and at the golden ratio the center still wins by a constant factor. The proof rests on a local limit theorem at the center, which holds for every first run that is finite almost surely. For runs of infinite mean we also prove a local form of Lamperti's law at the center. We also study an aging walk (the coin-turning walk), which switches at step k with probability c/k. With a fixed first step, every interior position ties with the end exactly at $c = 1$, for every N. The same holds for an aging walk with d states, whose vector of counts is exactly uniform at the tie.

Keywords. Lévy walk, alternating renewal process, local limit theorem, persistent random walk, coin-turning walk, Pólya urn.
Mathematics Subject Classification (2020). Primary 60G50, 60K05. Secondary 60F05, 60J10, 60C05, 82C41, 82C70.

1 Introduction

We study walks on $\mathbb{Z}$ that move in runs of alternating direction with random lengths. Let A_N be the number of right steps among the first N. If the run lengths are independent, A_N is the occupation time of an alternating renewal process [7, 8]. If they are also heavy-tailed, the walk is a lattice analogue of a Lévy walk. Lévy walks are a standard model of superdiffusion [10], and they model anomalous heat transport in one-dimensional lattices [11, 15]. The profile of a Lévy walk has a central peak and two ballistic side peaks [13, 9, 10], and the part of the profile between them is described by an infinite density [28, 29]. On the lattice we compare the end bins $A_N = N$ and $A_N = 0$, which the walk reaches only if one run covers all N steps, with the center $A_N = N/2$, which it reaches in many ways. In the picture of heat spreading, the end bins are the lattice form of the ballistic side peaks of the standard Lévy walk (without velocity fluctuations, see Remark 2.6), the center bin is the top of the central peak, and the stationary start used below is the equilibrated initial ensemble [9]. We ask when the ballistic ends beat the center. For runs with

geometric lengths this question is Problem 131 of Kagey's Open Problem Collection [1], posed for a Galton board. The walk is then the persistent random walk [38, 39], which remembers only its last step, and the center catches up with the ends when the expected number of switches is about $\log N$ [2].

In this paper we ask what happens if the walk remembers more. We study two kinds of memory.

- A walk with heavy-tailed runs remembers how long it has gone in its current direction. Its path is a sequence of runs of alternating direction, and the run lengths ξ are independent with $\mathbb{P}(\xi \geq k) = k^{-\alpha}$ for $k \geq 1$.
- An aging walk remembers how old it is. It switches at step $k \geq 2$ with probability c/k. This is the coin-turning walk of Engländer and Volkov [24, 32] with $p_k = c/k$. (In the literature on renewal processes and Lévy walks, aging means instead that the process depends on the time since it was prepared [7, Sections 8 and 9].) We also give the walk d directions. Then at step $k \geq 2$ it turns to each of the other $d-1$ directions with probability $\theta/(k+d-2)$, which is c/k with $c = \theta$ when $d = 2$.

The answer depends on the kind of memory. For heavy-tailed runs started in equilibrium the ends win for large N if α is below the golden ratio $\varphi = (1+\sqrt{5})/2$, and the center wins if α is above φ. With this start the end decays like $N^{1-\alpha}$ (Proposition 2.1) and the center like $N^{-1/\alpha}$ (Theorem 2.4), and the two exponents are equal exactly when $\alpha^2 = \alpha + 1$. More precisely, the ratio of end to center is asymptotic to $C(\alpha)N^{1-\alpha+1/\alpha}$ with an explicit constant $C(\alpha)$ (Corollary 2.5).

For the aging walk with a fixed first step, every bin from 1 to $N-1$ ties with the end exactly at $c = 1$, for every $N \geq 2$. With d directions the same happens at $\theta = 1$, where the vector of counts is exactly uniform (Section 3). At $c = 1$ the walk makes $H_N - 1$ switches on average, which is about $\log N$, as at the crossing for geometric runs. Earlier work on the aging walk gives the limit law of A_N/N [23, 24] and the scaling limit of the walk [32], and the uniform law for two directions is an exercise in [32]. Its joint law with the last step is in the book of Engländer and Volkov [31], as we explain in Section 1.4. In both models the start matters. For heavy-tailed runs it decides whether the threshold is φ or 1, and for the aging walk it decides whether the crossing is exactly $c = 1$. The main results are Theorem 2.4 and Corollary 2.5 for heavy-tailed runs and Proposition 3.3 and Theorem 3.4 for aging walks. Section 1.3 lists all the results.

1.1 Why the golden ratio

Here is the heuristic. Theorem 2.4 and Corollary 2.5 make it rigorous. For heavy-tailed runs take $1 < \alpha < 2$, so a run has a finite mean μ but an infinite variance. Start the walk in equilibrium, at a typical time of a long sequence of runs. Then the first run is already in progress, and long runs are more likely to be in progress. The chance that this run covers all N steps is of order $N^{1-\alpha}$, and this is the end. For the center, the runs to the right and the runs to the left form two independent renewal processes, each making about $N/(2\mu)$ runs, and the walk ends in the center when they meet at level $N/2$. Each process fluctuates on the scale $N^{1/\alpha}$ of a stable law, so they meet at the same level with probability of order $N^{-1/\alpha}$. Hence the ratio of end to center is of order $N^{1-\alpha}/N^{-1/\alpha} = N^{1-\alpha+1/\alpha}$. The exponent vanishes when $\alpha^2 - \alpha - 1 = 0$, that is, at $\alpha = \varphi$. Below φ the ends win, and above φ the center wins. If the first run is a fresh run instead, the end is of order $N^{-\alpha}$, a factor N smaller, and the center wins for every $1 < \alpha < 2$.

1.2 Setting and notation

Every walk in this paper takes N steps $X_1, \dots, X_N \in \{+1, -1\}$, where $+1$ means right. Let $A_N = \#\{i \le N : X_i = +1\}$ be the number of right steps, and write $P_N(k) = \mathbb{P}(A_N = k)$ for the probability of bin k, that is, of k right steps, for whichever walk we study. Unless we say otherwise the first step is fair, $\mathbb{P}(X_1 = +1) = \frac{1}{2}$, and the rules treat the two directions alike, so $P_N(k) = P_N(N-k)$. We write $\mathbb{P}_+$ and $\mathbb{E}_+$ for probability and expectation given $X_1 = +1$. For even N we put $h = N/2$ and call $C_N = P_N(h)$ the center and $E_N = P_N(N) = P_N(0)$ the end. Whenever C_N appears, N is even. A crossing is a value of the parameter at which a bin ties with an end, and for the center this means $C_N = E_N$. For heavy-tailed runs we do not locate a crossing at fixed N. There we prove which of C_N and E_N is larger for all large N at fixed α, and the golden ratio is the threshold for this sign. All logarithms are natural, $H_k = \sum_{j \le k} 1/j$ is the k-th harmonic number, and γ is Euler's constant. Limits, $O(\cdot)$ and $o(\cdot)$ refer to $N \to \infty$, over even N when C_N appears. The constants implied by $O(\cdot)$ are absolute unless we name what they depend on.

Heavy-tailed runs (Section 2) have the exponent α, the mean run length $\mu = \mathbb{E}\xi = \zeta(\alpha)$ for $\alpha > 1$, and the length D of the first run. A fresh start has D distributed as ξ, and the stationary start has $\mathbb{P}(D = \ell) = \mathbb{P}(\xi \ge \ell)/\mu$. We put $R_N = E_N/C_N$, so the ends win exactly when $R_N > 1$. The aging walk (Section 3) has the constant $c \in (0, 2)$, and its version with d states has the constant θ and the count vector $(n_1, \dots, n_d)$, where n_i is the number of steps in state i. Symbols that only one section uses are defined in that section.

1.3 Results

Heavy-tailed runs. For $1 < \alpha < 2$ and every first run that is finite almost surely, C_N is asymptotic to an explicit constant times $N^{-1/\alpha}$ (Theorem 2.4). With the stationary start $R_N \sim C(\alpha) N^{1-\alpha+1/\alpha}$ for an explicit constant $C(\alpha)$, and $C(\varphi) = 0.7761\dots$ (and $C(\varphi) < 0.952$ by elementary bounds), so at $\alpha = \varphi$ the center still wins. With a fresh start $R_N \to 0$ (Corollary 2.5). For $\alpha \ge 2$ the center is of order $N^{-1/2}$, with an extra factor $(\log N)^{-1/2}$ at $\alpha = 2$, and $R_N \to 0$ with either start (Theorem 2.8). For $0 < \alpha < 1$ the runs have infinite mean and only the fresh start makes sense. Then $NC_N \to \frac{2}{\pi} \tan \frac{\pi\alpha}{2}$ and $R_N \to \infty$ (Theorem 2.9). So with a fresh start the ends win for $\alpha < 1$ and the center wins for $\alpha > 1$. We do not prove what happens at $\alpha = 1$ (Remark 2.10).

Aging. With a fixed first step and $c = 1$ the number of right steps is exactly uniform on $\{1, \dots, N\}$. The ratio of each interior bin to the end increases strictly in c, so every interior bin ties with the end exactly at $c = 1$, for every $N \ge 2$ (Proposition 3.3). With d states, a start in state 1 and $\theta = 1$, the count vector is exactly uniform over all compositions $(n_1, \dots, n_d)$ of N with $n_1 \ge 1$, and every other such bin ties with the corner $(N, 0, \dots, 0)$ exactly at $\theta = 1$, for every $N \ge 2$ (Theorem 3.4). A small change of the aging rate gives a chain that redraws or repeats its state, and its counts have exactly the law of Pólya's urn for every weight (Proposition 3.1). With a fair first step each interior bin is twice as likely as an end at $c = 1$, and its crossing is unique and below 1 (Proposition 3.6). For the center the crossing is $c_N = 1 - \log 2/(\log N + \gamma + 2 - 2\log 2) + O((\log N)^{-3})$ (Proposition 3.7).

Methods. For heavy-tailed runs the walk is made of two independent renewal processes, one for the runs of each sign (Proposition 2.2). At the center we need a local limit theorem for each of them, with a delay for the first run. We obtain it from Gnedenko's local limit theorem for the

sums of the run lengths. Sums of few runs that lie far above their mean are controlled by local bounds in the big-jump domain. For $\alpha < 1$ these bounds need one more step, in which we cut the runs into two halves. Then a Mellin transform identifies the limit with Lamperti's density. For the aging walk all the exact laws come from one chain that draws afresh or repeats its state. The ties come from writing each ratio of a bin to the end as a sum over paths of products of odds that increase in c. For the fair start we couple the walk with a sign process that switches at the points of a Poisson process, and summation by parts turns the coupling into a local limit theorem with a rate.

1.4 Prior work

For heavy-tailed runs, Godrèche and Luck [7] give the transform of the occupation time of an alternating renewal process started at a renewal. They give the decay of the end for both starts [7, Eqs. (1.6) and (8.7)], and from the transform they derive, by a formal Laplace analysis in continuous time, the stable law of the occupation time for $1 < \alpha < 2$ [7, Section 7(iii)]. Read at the center, that law gives the constant of Theorem 2.4. In probability theory the walk is the one-dimensional directionally reinforced random walk of Mauldin, Monticino and von Weizsäcker [35], whose limit laws were found by Horváth and Shao [36]. Cénac, Le Ny, de Loynes and Offret study it as a persistent random walk whose runs have general lengths. They give criteria for recurrence [33] and prove functional scaling limits [34], a Brownian motion for $\alpha \geq 2$, a stable Lévy process for $1 < \alpha < 2$ and an arcsine Lamperti process for $\alpha < 1$. Theorems 2.4, 2.8 and 2.9 are local forms of the marginals of these limits at the center. The end E_N is a persistence event, and it is not a local form of these limits. Magdziarz, Scheffler, Straka and Żebrowski [37] prove limit theorems for Lévy walks in continuous space and time. Zaburdaev, Denisov and Klafter [10] and Dhar, Saito and Derrida [11] describe the profile of a Lévy walk. The scalings of the central peak and of the side peaks were known before [12, 13, 9, 10], but we did not find the balance of the two stated there. The observation that equal decay of the two singles out the golden ratio is due, as far as we know, to Wang, Li and Hänggi [15, Section 6.2.1.3, Eq. (6.49)], who used it as a heuristic for the exponent of anomalous heat conduction. They also mention that Lee-Dadswell, Nickel and Gray [16] had found the same conduction exponent $(3 - \sqrt{5})/2$ by a hydrodynamic argument. The two decay rates of the end, $N^{1-\alpha}$ and $N^{-\alpha}$, are the scalings of the ballistic peaks of a Lévy walk with and without an equilibrated start [9, Table I]. What we add for heavy-tailed runs is the following. We prove a lattice local limit theorem at the center for every first run that is finite almost surely, so for both starts. With it the comparison of the end with the center becomes a theorem with the explicit constant $C(\alpha)$. At $\alpha = \varphi$ the center wins strictly, since $C(\varphi) < 1$. On the lattice the end and the center are both probabilities of bins, so the comparison of a weight with a height in [15] becomes exact. The comparison concerns the unbroadened peaks, and with velocity fluctuations the balance moves away from the golden ratio (Remark 2.6). For $\alpha < 1$ the limit law of A_N/N is Lamperti's [8], and we prove its local form at the center. We found only the integral law in the literature.

For aging walks, Dietz and Sethuraman [23] proved the limit law of A_N/N, Engländer and Volkov [24] gave a second proof, and Engländer, Volkov and Wang [32] found the scaling limit of the walk. For d states the limit is a Dirichlet law [23, Theorem 1.4], and Bouguet and Cloez [26, Propositions 4.1 and 4.5] find the joint limit of the fractions of time and the current state. The exact uniform law for two states is an exercise in [32, Section 3.1, footnote 1], which also announces a general connection with Pólya urns. Dietz, Lippitt and Sethuraman [27, Theorem 2.18] write the moments of these limit laws as occupation probabilities at finite N of a related chain. For the

chain that redraws or repeats its state (Proposition 3.1), which is a conservative random walk in the sense of [30], this gives the law of the counts, and at $\theta = 1$ the uniform law of Theorem 3.4(a). Engländer and Volkov work out the connection with urns in Chapter 7 of their book [31]. For two states they give the joint law of the counts and the last state in the case that becomes our $c = 1$ after a shift of one step [31, Claim 7.1]. For an integer weight they show that the counts and the last state have the law of Pólya's urn at every time, which is Proposition 3.1 for $d = 2$ and an integer θ [31, Theorem 7.2]. They credit this result to E. Crane, and they call the relation mimicking [31, Definition 7.1 and Lemma 7.1]. The book also gives exact laws for a fair first step [31, Section 5.7]. Pradhan, Roy and Ray [40] study, with the methods of physics, a persistent walk whose probability of reversal decays like a power of the time. When the power is one it is close to our aging walk, and they find its second moment and the tail of its runs. What we add in Section 3 is the joint law of the counts and the last state for $d \geq 3$ states or a weight θ that is not an integer, the strict ordering that turns the uniform laws into ties (Proposition 3.3(b) and Theorem 3.4(b)), and for the fair start the expansion of the crossing (Proposition 3.7) with the local limit theorem with a rate behind it (Proposition B.5).

1.5 Organization

Section 2 treats heavy-tailed runs, first the golden ratio for $1 < \alpha < 2$ and then the exponents $\alpha \geq 2$ and $\alpha < 1$. Section 3 treats the aging walk, first the chain that draws afresh or repeats, then a fixed first step, d states and a fair first step. Section 4 describes the checks. The longer proofs are in the appendices. Appendix A holds the proofs for Section 2, and Appendix B proves Proposition 3.7.

2 Heavy-tailed runs

In this section the walk remembers the time since its last switch. The steps form maximal runs whose directions alternate. The first run has length D, and the later runs have i.i.d. lengths $\xi_1, \xi_2, \dots$ distributed as ξ, where

$$\bar{F}(k) = \mathbb{P}(\xi \geq k) = k^{-\alpha} \quad (k \geq 1)$$

for some $\alpha > 0$. So $\mathbb{P}(\xi = k) = k^{-\alpha} - (k+1)^{-\alpha} > 0$ for every $k \geq 1$. The first direction is fair, and D, (ξ_i) and the first direction are independent. In probability theory this is the one-dimensional directionally reinforced random walk [35, 36], and a persistent random walk in the sense of [33, 34]. In physics it is a lattice Lévy walk. With a fresh start D has the law of ξ. For $\alpha > 1$ the stationary start is $\mathbb{P}(D = \ell) = \bar{F}(\ell)/\mu$, where $\mu = \mathbb{E}\xi = \zeta(\alpha)$. This law is stationary in the following sense. The number of steps left in the current run, counting the current one, is a Markov chain that moves from $k \geq 2$ to $k - 1$ and from 1 to a fresh copy of ξ. Since $\bar{F}(k) = \bar{F}(k+1) + \mathbb{P}(\xi = k)\bar{F}(1)$, the law $\bar{F}(k)/\mu$ is invariant for this chain. So the stationary start sees the runs at a typical time. We define A_N, P_N, C_N and E_N as in Section 1.2, and put $R_N = E_N/C_N$. In this section the constants implied by $O(\cdot)$ may depend on α.

Proposition 2.1 (the end)**.** *For every law of D and every $N \geq 1$, $E_N = \frac{1}{2}\mathbb{P}(D \geq N)$. With a fresh start $E_N = \frac{1}{2}N^{-\alpha}$. With the stationary start*

$$E_N = \frac{\zeta(\alpha, N)}{2\zeta(\alpha)} = \frac{N^{1-\alpha}}{2(\alpha-1)\zeta(\alpha)}\Big(1 + \frac{\vartheta_N(\alpha-1)}{N}\Big)$$

for some $\vartheta_N \in [0, 1]$, where $\zeta(\alpha, N) = \sum_{k \geq N} k^{-\alpha}$ is the Hurwitz zeta function.

Proof. All N steps are $+1$ exactly when the first direction is $+$ and $D \geq N$. Hence $E_N = \frac{1}{2}\mathbb{P}(D \geq N)$. For the fresh start $\mathbb{P}(D \geq N) = \bar{F}(N) = N^{-\alpha}$. For the stationary start $\mathbb{P}(D \geq N) = \sum_{\ell \geq N} \ell^{-\alpha}/\mu = \zeta(\alpha, N)/\zeta(\alpha)$. To estimate $\zeta(\alpha, N)$ we compare the sum with an integral. Since $t^{-\alpha}$ is decreasing,

$$\int_N^\infty t^{-\alpha}\, dt \leq \zeta(\alpha, N) \leq N^{-\alpha} + \int_N^\infty t^{-\alpha}\, dt.$$

The integral equals $N^{1-\alpha}/(\alpha - 1)$, and $N^{-\alpha}$ is $(\alpha - 1)/N$ times the integral. This gives the second form. □

So the end only sees the first run, while the center depends on all the runs. For the center we need the renewal quantities

$$\tau_m = \xi_1 + \cdots + \xi_m, \quad \tau_0 = 0, \qquad u_m(k) = \mathbb{P}(\tau_m = k),$$
$$q_m(k) = \mathbb{P}(\tau_{m-1} < k \leq \tau_m) = \sum_{j \geq 1} \bar{F}(j) u_{m-1}(k - j)$$

for $m, k \geq 1$. So $u_m(k)$ is the chance that the m-th run ends at level k, and $q_m(k)$ the chance that the m-th run covers level k. For the delayed versions let D be independent of (ξ_i), and put $u_m^D(k) = \mathbb{P}(D + \tau_{m-1} = k)$, $q_1^D(k) = \mathbb{P}(D \geq k)$ and $q_m^D(k) = \mathbb{P}(D + \tau_{m-2} < k \leq D + \tau_{m-1})$ for $m \geq 2$. For the stationary start $u_m^D(k) = \sum_\ell \frac{\bar{F}(\ell)}{\mu} u_{m-1}(k - \ell) = q_m(k)/\mu$.

The $+$ runs and the $-$ runs form two independent renewal processes. For $1 \leq a \leq N - 1$ the walk is in bin a at time N exactly when one process has a run that ends exactly at its level (a for the $+$ runs and $N - a$ for the $-$ runs) and the other process has a run that covers its level.

Proposition 2.2 (two renewal processes)**.** *Let D have any law on $\{1, 2, \dots\}$. For $a + b = N$ with $a, b \geq 1$,*

$$\mathbb{P}_+(A_N = a) = \sum_{m \geq 1} \Big[u_m(b)\, q_{m+1}^D(a) + u_m^D(a)\, q_m(b) \Big],$$

and $\mathbb{P}_+(A_N = N) = \mathbb{P}(D \geq N)$. So for even N and $h = N/2$, $C_N = \sum_{m \geq 1} [u_m(h) q_{m+1}^D(h) + u_m^D(h) q_m(h)]$. For the fresh start this is $C_N = \sum_m u_m(h)[q_m(h) + q_{m+1}(h)]$.

Proof. We split according to the run that contains step N. Let the $+$ runs have lengths $D, \xi_2^+, \xi_3^+, \dots$ and the $-$ runs $\xi_1^-, \xi_2^-, \dots$, and let $s_j^\pm$ be the sum of the first j lengths of each sign, with $s_0^\pm = 0$. So s_j^+ has law u_j^D and s_j^- has law u_j, and the two sequences are independent. The runs alternate and the first one is a $+$ run. So the j-th $+$ run comes after $j - 1$ runs of each sign and covers the steps in $(s_{j-1}^+ + s_{j-1}^-,\, s_j^+ + s_{j-1}^-]$, and the j-th $-$ run covers $(s_j^+ + s_{j-1}^-,\, s_j^+ + s_j^-]$. Let $N = a + b$. There are three cases.

Suppose step N is in the first run. Then $A_N = N$, and this happens exactly when $D \geq N$.

Suppose step N is in the j-th $+$ run with $j \geq 2$. Then exactly $j - 1$ of the $-$ runs lie before step N, and they are complete, so $A_N = N - s_{j-1}^-$. So $A_N = a$ means $s_{j-1}^- = b$, and then step $N = a + b$ is in this run exactly when $s_{j-1}^+ < a \leq s_j^+$. By independence this has probability $u_{j-1}(b)\, q_j^D(a)$.

Suppose step N is in the j-th $-$ run. Then the first j of the $+$ runs are complete and no other has begun, so $A_N = s_j^+$. So $A_N = a$ means $s_j^+ = a$, and then step N is in this run exactly when $s_{j-1}^- < b \leq s_j^-$. This has probability $u_j^D(a)\, q_j(b)$.

The cases are disjoint. Summing them, with $m = j - 1$ in the second and $m = j$ in the third, gives the formula. Now let N be even. The first direction is fair, and changing the sign of every

step sends A_N to $N - A_N$. Hence $C_N = \frac{1}{2}[\mathbb{P}_+(A_N = h) + \mathbb{P}_+(A_N = N - h)] = \mathbb{P}_+(A_N = h)$. With a fresh start $u_m^D = u_m$ and $q_m^D = q_m$, which gives the last formula. □

For a fresh start this identity is the coefficient form of the double transform of Godrèche and Luck [7, Section 7] for the occupation time of an alternating renewal process. A general first run only changes the factors u_m^D and q_{m+1}^D. To evaluate it we need the stable limit of τ_m and its local form.

Lemma 2.3 (stable and local limits)**.** *Let $1 < \alpha < 2$ and $a_m = m^{1/\alpha}$. Put $K_\alpha = \Gamma(2-\alpha)/(\alpha-1) = -\Gamma(1-\alpha) > 0$ and $\sigma_\alpha = 2K_\alpha|\cos\frac{\pi\alpha}{2}|$.*

(i) $\mathbb{E}e^{i\theta\xi} = 1 + i\mu\theta + K_\alpha(-i\theta)^\alpha + O(\theta^2)$ as $\theta \to 0$, with the principal branch.

(ii) $(\tau_m - m\mu)/a_m$ converges in law to a variable G with $\mathbb{E}e^{-sG} = e^{K_\alpha s^\alpha}$ for $s \ge 0$ and $\mathbb{E}e^{i\theta G} = \exp(K_\alpha(-i\theta)^\alpha)$. The law of G is totally skewed to the right (skewness $\beta = 1$ and scale $(K_\alpha|\cos\frac{\pi\alpha}{2}|)^{1/\alpha}$ in the usual parametrization) and has a bounded, uniformly continuous density g with $g(z) \to 0$ as $|z| \to \infty$.

(iii) (Gnedenko) The constant $\bar{u} := \sup_{m\ge1}\sup_k a_m u_m(k)$ is finite, and $a_m u_m(k) - g((k - m\mu)/a_m) \to 0$ uniformly in k as $m \to \infty$.

(iv) If G' is an independent copy of G, then $G - G'$ is symmetric α-stable with $\mathbb{E}e^{i\theta(G-G')} = e^{-\sigma_\alpha|\theta|^\alpha}$, and $I_\alpha := \int_{\mathbb{R}} g(z)^2\,dz = \Gamma(1 + 1/\alpha)/(\pi\sigma_\alpha^{1/\alpha})$.

Proof. All four parts come from the expansion of the transform of ξ near 0. We find it first.

(i) Summation by parts gives $1 - \mathbb{E}w^\xi = (1-w)\sum_{k\ge1}\bar{F}(k)w^{k-1}$ for $|w| \le 1$. Put $w = e^{-v}$ with $\operatorname{Re} v \ge 0$, and write $\operatorname{Li}_\alpha(w) = \sum_{k\ge1} k^{-\alpha}w^k$. Since $\bar{F}(k) = k^{-\alpha}$, the identity reads $1 - \mathbb{E}e^{-v\xi} = (e^v - 1)\operatorname{Li}_\alpha(e^{-v})$. For $0 < |v| < 2\pi$, [22, 25.12.12] gives $\operatorname{Li}_\alpha(e^{-v}) = \Gamma(1-\alpha)v^{\alpha-1} + \zeta(\alpha) + O(|v|)$. We multiply by $e^v - 1 = v + O(|v|^2)$ and use $\zeta(\alpha) = \mu$ and $\Gamma(1-\alpha) = -K_\alpha$. Since $\alpha + 1 > 2$,

$$1 - \mathbb{E}e^{-v\xi} = \mu v - K_\alpha v^\alpha + O(|v|^2) \qquad (v \to 0,\ \operatorname{Re} v \ge 0).$$

With $v = -i\theta$ this is (i).

(ii) Put $Z_m = (\tau_m - m\mu)/a_m$. Fix θ and let $t = \theta/a_m$. By (i) and $e^{-i\mu t} = 1 - i\mu t + O(t^2)$, and since $a_m^\alpha = m$,

$$e^{-i\mu t}\,\mathbb{E}e^{it\xi} = 1 + K_\alpha(-it)^\alpha + O(t^2) = 1 + \frac{K_\alpha(-i\theta)^\alpha}{m} + O\big(\theta^2 m^{-2/\alpha}\big).$$

The left side is the characteristic function of $(\xi - \mu)/a_m$ at θ, and $\mathbb{E}e^{i\theta Z_m}$ is its m-th power. As $2/\alpha > 1$, the error is $o(1/m)$, so $\mathbb{E}e^{i\theta Z_m} \to \exp(K_\alpha(-i\theta)^\alpha)$. The real part of $(-i\theta)^\alpha$ is $|\theta|^\alpha\cos\frac{\pi\alpha}{2}$, which is negative because $1 < \alpha < 2$. So the limit has modulus $\exp(-\frac{1}{2}\sigma_\alpha|\theta|^\alpha)$. It is continuous at 0, so by Lévy's continuity theorem it is the characteristic function of a variable G, and Z_m converges in law to G. It is integrable, so by Fourier inversion G has a bounded, uniformly continuous density g, and g vanishes at infinity by the Riemann–Lebesgue lemma (the domain of attraction also follows from [20, Section XVII.5]). Since $\cos\frac{\pi\alpha}{2} < 0$, we can write $K_\alpha(-i\theta)^\alpha = -K_\alpha|\cos\frac{\pi\alpha}{2}|\,|\theta|^\alpha\big(1 - i\,\operatorname{sign}(\theta)\tan\frac{\pi\alpha}{2}\big)$. So in the usual parametrization of stable laws G has skewness $\beta = 1$ and scale $(K_\alpha|\cos\frac{\pi\alpha}{2}|)^{1/\alpha}$, and this is what totally skewed to the right means.

For the Laplace transform let $s > 0$ be real and take $v = s/a_m$ in the expansion above. In the same way $\mathbb{E}e^{-sZ_m} \to e^{K_\alpha s^\alpha}$. This also holds with $2s$ in place of s, so the second moments of

e^{-sZ_m} are bounded in m, and these variables are uniformly integrable. Hence $\mathbb{E}e^{-sZ_m} \to \mathbb{E}e^{-sG}$, and so $\mathbb{E}e^{-sG} = e^{K_\alpha s^\alpha}$.

(iii) Every $\mathbb{P}(\xi = k)$ is positive, so ξ has maximal span 1. So Gnedenko's local limit theorem [17, Theorem 4.2.1] applies to the convergence in (ii) and gives the uniform limit. For $\bar{u}$, the limit gives an m_0 with $a_m u_m(k) \le \|g\|_\infty + 1$ for all $m \ge m_0$ and all k. For $m < m_0$ we use $u_m(k) \le 1$, so $a_m u_m(k) \le a_{m_0}$. Hence $\bar{u} < \infty$.

(iv) The characteristic function of $-G'$ is the complex conjugate of that of G. So $G - G'$ has the characteristic function $|\mathbb{E}e^{i\theta G}|^2 = e^{-\sigma_\alpha|\theta|^\alpha}$, which is that of a symmetric α-stable law. Plancherel's theorem gives

$$I_\alpha = \int_{\mathbb{R}} g(z)^2\,dz = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\sigma_\alpha|\theta|^\alpha}\,d\theta = \frac{1}{\pi}\int_0^\infty e^{-\sigma_\alpha\theta^\alpha}\,d\theta = \frac{\Gamma(1+1/\alpha)}{\pi\sigma_\alpha^{1/\alpha}},$$

where the last step is the substitution $x = \sigma_\alpha\theta^\alpha$. (Note that I_α is the density of $G - G'$ at 0. This is how it enters Theorem 2.4, where two independent renewal processes have to meet.) □

Theorem 2.4 (the center for $1 < \alpha < 2$)**.** *Let $1 < \alpha < 2$ and let D be any a.s. finite first run, in particular either start. As $N \to \infty$ over even N,*

$$C_N \sim 2I_\alpha\Big(\frac{\mu}{h}\Big)^{1/\alpha} = \frac{2\Gamma(1+1/\alpha)}{\pi}\Big(\frac{\mu}{K_\alpha|\cos(\pi\alpha/2)|\,N}\Big)^{1/\alpha}.$$

In words, the center decays like $N^{-1/\alpha}$, and the first run does not affect the leading term. The constant agrees with the formal prediction of Godrèche and Luck [7, Eqs. (7.12) to (7.15)], who work in continuous time with a start at a renewal. Their stable law for the occupation time, read at the center with lattice spacing two, gives the constant above. The theorem is a local form, at the center, of the marginal of the functional limit of Cénac et al. [34, Theorem 3.2].

Idea of the proof. By Proposition 2.2 the walk ends in the center when, for some m, one renewal process ends its m-th run exactly at level h and the other is inside a run that covers level h. Level h takes about h/μ runs. Put $b_h = (h/\mu)^{1/\alpha}$, which is the scale on which τ_m spreads around its mean for such m. So the m-th run ends at h with probability about $g(z)/b_h$, where $z = (h - m\mu)/b_h$ says how far m is from the typical number of runs. A level is covered by a run μ times as often as a run ends there, so the m-th run covers h with probability about $\mu g(z)/b_h$. The values of m that matter have $|m\mu - h|$ of order b_h, and there are b_h/μ of them for each unit of z. Proposition 2.2 has two terms of this kind for each m, so C_N is about $2 \cdot \frac{1}{b_h} \cdot \frac{\mu}{b_h} \cdot \frac{b_h}{\mu}\int g^2 = 2I_\alpha/b_h$.

In the proof (Appendix A) we split the sum in Proposition 2.2 by the number m of runs. The terms with $|m\mu - h| \le Mb_h$ form a Riemann sum for $2b_h^{-1}\int_{-M}^{M} g^2$ by Lemma 2.3(iii). The terms with $m \le h/(4\mu)$ add up to $O(h^{-\alpha}) = o(b_h^{-1})$, since a sum of few runs that is far above its mean is most likely made by one long run. The others add up to at most $O(b_h^{-1})\mathbb{P}(|G| \ge M - 1) + o(b_h^{-1})$, and we let $M \to \infty$.

Corollary 2.5 (the golden-ratio exponent)**.** *Let $1 < \alpha < 2$ and $e(\alpha) = 1 - \alpha + 1/\alpha$. If $\mathbb{P}(D \ge N) \sim c_D N^{-\beta_D}$ with $c_D, \beta_D > 0$, then $R_N \sim c_D N^{1/\alpha-\beta_D}/(4I_\alpha(2\mu)^{1/\alpha})$. In particular $R_N \sim C(\alpha)N^{e(\alpha)}$ for the stationary start and $R_N \sim C_{\mathrm{fr}}(\alpha)N^{e(\alpha)-1}$ for the fresh start, where*

$$C(\alpha) = \frac{\pi\big(K_\alpha|\cos\frac{\pi\alpha}{2}|\big)^{1/\alpha}}{4(\alpha-1)\Gamma(1+\frac{1}{\alpha})\,\zeta(\alpha)^{1+1/\alpha}}, \qquad C_{\mathrm{fr}}(\alpha) = (\alpha-1)\zeta(\alpha)\,C(\alpha).$$

α	$e(\alpha)$	$C(\alpha)$	$C_{\mathrm{fr}}(\alpha)$	R_{1600}	R_{10^4}	R_{10^5}
1.3	+0.469231	0.422717	0.498631	17.7593 [1.3180]	38.7537 [1.2171]	106.149 [1.1317]
1.5	+0.166667	0.647971	0.846371	2.14409 [0.9675]	2.95280 [0.9818]	4.37642 [0.9914]
φ	0	0.776132	1.073675	0.661763 [0.8526]	0.697331 [0.8985]	0.727689 [0.9376]
5/3	$-1/15$	0.835101	1.182237	0.411815 [0.8064]	0.389307 [0.8614]	0.352822 [0.9102]
1.7	-0.111765	0.880046	1.265508	0.298369 [0.7733]	0.262005 [0.8334]	0.215839 [0.8881]
1.9	-0.373684	1.438986	2.266075	0.0449706 [0.4923]	0.0256505 [0.5569]	0.0121612 [0.6242]

Table 1: Constants of Corollary 2.5 and computed values of $R_N = E_N/C_N$ for the stationary start, with $R_N/(C(\alpha)N^{e(\alpha)})$ in brackets.

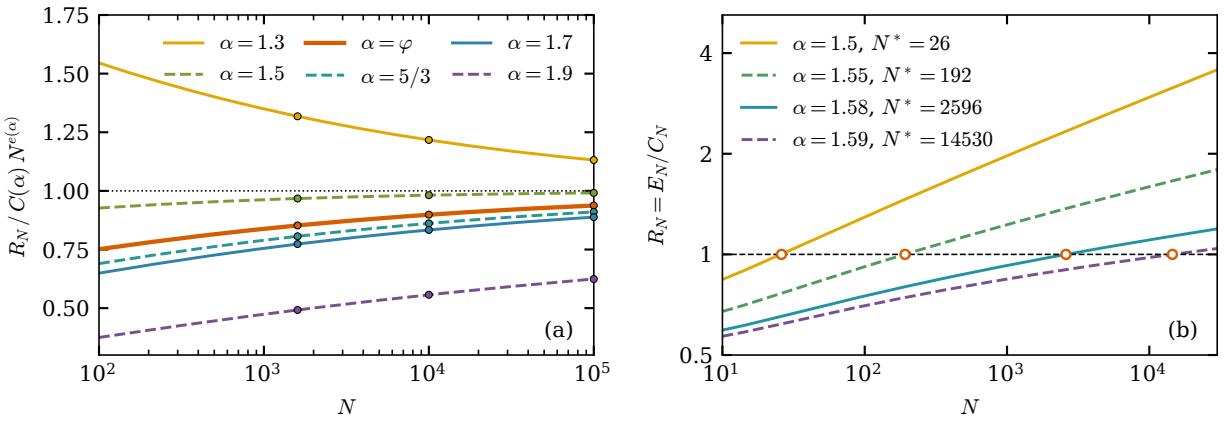


Figure 1: Alternating runs with a stationary start. (a) $R_N/(C(\alpha)N^{e(\alpha)})$ for even N from 100 to 10^5, with the entries of Table 1 as dots. (b) R_N for $\alpha = 1.5, 1.55, 1.58, 1.59$ below φ. Each curve crosses 1 once on $10 \le N \le 3 \cdot 10^4$, at $N^*(\alpha)$ (circles).

Since $e(\alpha) > 0$ exactly when $\alpha < \varphi$, with a stationary start the ends win for all large N if $\alpha < \varphi$, and the center wins if $\alpha > \varphi$. At $\alpha = \varphi$,

$$R_N \to C(\varphi) = \frac{\pi\varphi\big(\varphi\,\Gamma(\varphi^{-2})\big|\cos\tfrac{\pi\varphi}{2}\big|\big)^{1/\varphi}}{4\,\Gamma(\varphi)\,\zeta(\varphi)^{\varphi}} = 0.7761317206\ldots,$$

so the center still wins, by the factor $1/C(\varphi) = 1.2884\ldots$ in the limit. That $C(\varphi) < 1$ needs no computation (see the proof). With a fresh start $e(\alpha) - 1 < 0$, and the center wins for every $1 < \alpha < 2$.

Proof. By Proposition 2.1 and Theorem 2.4, with $h = N/2$, we have $R_N = \mathbb{P}(D \ge N)/(2C_N)$, and this is asymptotic to $c_D N^{-\beta_D}/(4I_\alpha(2\mu/N)^{1/\alpha})$, which is the general formula. By Lemma 2.3(iv), $4I_\alpha(2\mu)^{1/\alpha} = 4\Gamma(1+\frac{1}{\alpha})\mu^{1/\alpha}/(\pi(K_\alpha|\cos\frac{\pi\alpha}{2}|)^{1/\alpha})$. The fresh start has $c_D = 1$ and $\beta_D = \alpha$, which gives $C_{\mathrm{fr}}(\alpha)$. By Proposition 2.1 the stationary start has $c_D = 1/((\alpha-1)\mu)$ and $\beta_D = \alpha - 1$, which gives $C(\alpha)$. Next, $e(\alpha) = -(\alpha^2 - \alpha - 1)/\alpha$, and φ is the positive root of $\alpha^2 - \alpha - 1$. At $\alpha = \varphi$ we have $\alpha - 1 = 1/\varphi$, $1 + 1/\alpha = \varphi$ and $2 - \alpha = \varphi^{-2}$. So $K_\varphi = \varphi\,\Gamma(\varphi^{-2})$, and substituting these into $C(\alpha)$ gives the closed form. To see that $C(\varphi) < 1$, use four elementary bounds. Since $x^{-\varphi}$ is convex, the trapezoid rule overestimates its integral over $[1,\infty)$, so $\zeta(\varphi) - \frac{1}{2} \ge 1/(\varphi - 1) = \varphi$. The minimum of Γ on $(0,\infty)$ is $0.8856\ldots$ [22, Section 5.4(iii)], so $\Gamma(\varphi) \ge 0.8856$. Since $\Gamma \le 1$ on $[1,2]$, $\Gamma(\varphi^{-2}) = \Gamma(1+\varphi^{-2})\varphi^2 \le \varphi^2$. Finally $\frac{\pi}{2} < \frac{\pi\varphi}{2} < \frac{5\pi}{6}$, so $|\cos\frac{\pi\varphi}{2}| \le \frac{\sqrt{3}}{2}$. Together these give $C(\varphi) \le \pi\varphi(\sqrt{3}\varphi^3/2)^{1/\varphi}/(4 \cdot 0.8856\,(\varphi + \frac{1}{2})^{\varphi}) = 0.95136\ldots < 1$. □

α	start	center C_N	ratio R_N	status
$\alpha > 2$	stationary	$\sim \sqrt{2\mu/(\pi v N)}$	$\sim C_2(\alpha) N^{3/2-\alpha} \to 0$	Theorem 2.8(a)
$\alpha = 2$	stationary	$\sim \sqrt{2\mu/(\pi N \log N)}$	$\sim \sqrt{\pi \log N/(8\zeta(2)^3 N)} \to 0$	Theorem 2.8(b)
$1 < \alpha < 2$	stationary	$\sim 2I_\alpha(\mu/h)^{1/\alpha}$	$\sim C(\alpha) N^{e(\alpha)}$	Corollary 2.5
$1 < \alpha < 2$	fresh	$\sim 2I_\alpha(\mu/h)^{1/\alpha}$	$\sim C_{\mathrm{fr}}(\alpha) N^{e(\alpha)-1} \to 0$	Corollary 2.5
$\alpha = 1$	fresh	$\approx 2/(\pi^2 \tilde{m}_h)$	$\approx \pi^2 \tilde{m}_h/(8h) \sim \pi^2/(8 \log N)$	heuristic, not proved (Remark 2.10)
$0 < \alpha < 1$	fresh	$\sim \frac{2}{\pi} \tan \frac{\pi\alpha}{2} \cdot \frac{1}{N}$	$\sim \frac{\pi}{4 \tan(\pi\alpha/2)} N^{1-\alpha} \to \infty$	Theorem 2.9

Table 2: The center and the ratio $R_N = E_N/C_N$ for each exponent. The center wins when $R_N \to 0$ and the ends win when $R_N \to \infty$. For $\alpha \geq 2$ the center formula holds for every a.s. finite first run, and a fresh start also gives $R_N \to 0$.

Corollary 2.5 is the lattice form, with its constant, of the heuristic of [15, Section 6.2.1.3, Eq. (6.49)]. The approach to $C(\varphi)$ is slow. At $\alpha = \varphi$ we have $R_{10^5}/C(\varphi) = 0.938$, and R_N increases at every even N from 10 to 10^5 (Table 1 and Figure 1).

The golden ratio also appears for the elephant random walk with first step $+1$ and reversal probability ε. There the law of A_N is log-concave for every N exactly when $1 - 2\varepsilon \leq 1/\varphi$ [5, Proposition 2.2], and for $1 - 2\varepsilon > 0.80399$ it is not log-concave for arbitrarily large N [6, Theorem 4.4]. We know of no connection.

Remark 2.6 (broadened side peaks). The end bins correspond to the ballistic side peaks of the standard Lévy walk, in which every flight has the same speed. These peaks are δ-functions whose weight decays like $t^{1-\alpha}$ for the equilibrated start [9, Table I], with the tail exponent of the flight times in the role of α, while the central peak has height of order $t^{-1/\alpha}$ [15, Section 6.2.1.3, Eq. (6.49)]. With velocity fluctuations the side peaks broaden into humps. If the speed is redrawn at each collision, the humps have width of order t and height of order $t^{-\alpha}$, and if it fluctuates during each flight, they have width of order $t^{1/2}$ and height of order $t^{1/2-\alpha}$ [9, Table I]. Matching these heights with $t^{-1/\alpha}$ gives $\alpha = 1$ in the first case, so the center wins for every $1 < \alpha < 2$, and in the second case the root $\alpha = (1 + \sqrt{17})/4 = 1.28\ldots$ of $\alpha^2 - \alpha/2 - 1 = 0$. So in the second case the side humps are higher than the central peak exactly when $\alpha < (1 + \sqrt{17})/4$. The golden ratio belongs to the unbroadened peaks, which our lattice walk has.

Remark 2.7 (crossover sizes and heat conduction). For $\alpha < \varphi$ and a stationary start the crossover is late when α is close to φ. In our computation, for each $\alpha \in \{1.5, 1.55, 1.58, 1.59\}$, the sign of $R_N - 1$ changes exactly once as N runs over the even numbers in $[10, 3 \cdot 10^4]$, namely at $N^*(\alpha) = 26, 192, 2596, 14530$. Heuristically, $C(\alpha) N^{e(\alpha)} = 1$ gives $\log N^*(\alpha) = 0.1834/(\varphi - \alpha) + O(1)$ as $\alpha \uparrow \varphi$. The $O(1)$ term is not small. At $\alpha = 1.59$ the leading term gives $\log N^* \approx 6.5$, while the computed value is $\log 14530 \approx 9.6$. At the value $\alpha = 5/3$ of Lévy-walk models of heat conduction [12, 11, 10], $R_N \sim C(5/3) N^{-1/15}$ with $C(5/3) = 0.8351$.

We turn to the other exponents. Table 2 collects them. For $\alpha \geq 2$ the sums τ_m have a Gaussian limit, and the center wins with either start.

Theorem 2.8 ($\alpha \geq 2$). *Let D be any a.s. finite first run, and let $N \to \infty$ over even N.*

(a) Let $\alpha > 2$ and $v = \operatorname{Var} \xi = 2\zeta(\alpha - 1) - \zeta(\alpha) - \zeta(\alpha)^2$. Then $C_N \sim \sqrt{2\mu/(\pi v N)}$. With the stationary start $R_N \sim C_2(\alpha) N^{3/2-\alpha}$, where $C_2(\alpha) = \sqrt{\pi v/(2\mu)}/(2(\alpha - 1)\mu)$.

(b) Let $\alpha = 2$, so that $\mu = \zeta(2)$ and $\mathbb{E}\xi^2 = \infty$. Then $(\tau_m - m\mu)/\sqrt{m \log m}$ converges in law to the standard normal law, and $C_N \sim \sqrt{2\mu/(\pi N \log N)}$. With the stationary start $R_N \sim \sqrt{\pi \log N/(8\zeta(2)^3 N)}$.

In both cases $R_N \to 0$ with the fresh start too.

In words, for $\alpha \geq 2$ the center is of the Gaussian order $N^{-1/2}$, with an extra factor $(\log N)^{-1/2}$ at $\alpha = 2$, where the variance is infinite. With either start the end is at most of order $N^{1-\alpha}$, so the center wins. The exponent of R_N is continuous at $\alpha = 2$, since $e(2) = -\frac{1}{2} = \frac{3}{2} - 2$.

Idea of the proof. The proof of Theorem 2.4 uses the exponent α in three places only. These are the limit law of τ_m with its local form, the bound $u_m(k) \leq C_B m k^{-\alpha-1}$ for a sum far above its mean, and the fact that the norming sequence varies regularly with index less than 1. All three hold for $\alpha \geq 2$ with the normal law in place of the stable law. For $\alpha > 2$ the norming is $a_m = \sqrt{vm}$. For $\alpha = 2$ the truncated second moment $\mathbb{E}[\xi^2; \xi \leq x]$ grows like $2 \log x$, which gives the norming $a_m = \sqrt{m \log m}$. Since $\int g^2 = 1/(2\sqrt{\pi})$ for the standard normal density g, the center is $1/(\sqrt{\pi}\, b_h)$ to leading order, where b_h is the norming at $m = h/\mu$. We give the proof in Appendix A, where the proof of Theorem 2.4 is written so that it covers both theorems.

The convergence is slow at $\alpha = 2$. There R_N divided by the leading term in (b) is 0.885 at $N = 1600$ and 0.943 at $N = 10^5$. At $\alpha = 3$ the ratio of R_N to the leading term in (a) is 0.981 at $N = 16000$.

For $0 < \alpha < 1$ the runs have infinite mean, and only the fresh start makes sense. Then A_N/N has a limit law that is not concentrated at a point, namely Lamperti's occupation-time law [8, 7], with density

$$\ell_\alpha(y) = \frac{\sin \pi\alpha}{\pi} \frac{y^{\alpha-1}(1-y)^{\alpha-1}}{y^{2\alpha} + 2y^\alpha (1-y)^\alpha \cos \pi\alpha + (1-y)^{2\alpha}} \qquad (0 < y < 1).$$

So the center should be of order $1/N$. The next theorem is the local form of this at the center.

Theorem 2.9 $(0 < \alpha < 1)$**.** *Let $0 < \alpha < 1$, with a fresh start. As $N \to \infty$ over even N,*

$$NC_N \to \ell_\alpha(\tfrac{1}{2}) = \frac{2}{\pi} \tan \frac{\pi\alpha}{2}.$$

Hence $R_N \sim \frac{\pi}{4 \tan(\pi\alpha/2)} N^{1-\alpha} \to \infty$.

In words, for $\alpha < 1$ the center is an ordinary bin of size about $1/N$, while the end has the much larger probability $\frac{1}{2} N^{-\alpha}$ that the first run goes right and lasts all N steps. So the ends win for every $\alpha < 1$. For the continuum Lévy walk with $0 < \alpha < 1$, the ballistic regime, Froemberg et al. [14] find the asymptotic densities. For our walk the integral form of the theorem is the arcsine Lamperti marginal of the scaling limit of Cénac et al. [34, Theorem 3.3 and Eq. (3.25)], and the theorem is its local form at the center.

Idea of the proof. Now τ_m is of order $a_m = m^{1/\alpha}$ with no centering, and τ_m/a_m converges to a positive stable variable G with density g. The number m of runs of each sign is no longer concentrated. It ranges over all m with $m^{1/\alpha}$ of order h, and on this range the terms of Proposition 2.2 are $\frac{2}{\alpha h m} t^2 g(t)^2$ with $t = h/a_m$. Summing over m is integrating in $\log m$, which gives $NC_N \to 4 \int_0^\infty t g(t)^2 \, dt$. This integral is the density at $\frac{1}{2}$ of $G_1/(G_1 + G_2)$ for two independent copies of G, and a Mellin transform identifies the law of G_1/G_2 and hence this density. The proof is in Appendix A. It does not use Lamperti's theorem.

Here too the convergence is slow near $\alpha = 1$. At $N = 16000$ the ratio of NC_N to its limit is 1.042, 1.001, 0.928 and 0.554 for $\alpha = 0.3$, 0.5, 0.7 and 0.9. We do not prove the local limit at bins other than the center.

Remark 2.10 (the exponent $\alpha = 1$ and the fresh start). With a fresh start the threshold is $\alpha = 1$. The ends win for $\alpha < 1$ by Theorem 2.9, and the center wins for $\alpha > 1$ by Corollary 2.5 and Theorem 2.8. At $\alpha = 1$ we only have a heuristic. Suppose we repeat the proof of Theorem 2.4 with a skewed Cauchy limit, the centering $m(H_{m+1} - 1)$, and the density $1/\pi^2$ at 0 of the difference of two copies of the limit. This suggests the row of Table 2, namely $R_N \approx \pi^2 \tilde{m}_h/(8h) \sim \pi^2/(8 \log N)$, where $\tilde{m}_h$ solves $m(H_{m+1} - 1) = h$. We have not controlled the error terms, so the claim that the center wins by a logarithm at $\alpha = 1$ is not proved. The computed R_N divided by $\pi^2 \tilde{m}_h/(8h)$ is 0.796, 0.850 and 0.886 at $N = 1600$, 10^4 and 10^5.

We do not know a rate in Theorem 2.4, whether R_N is monotone with a single sign change of $R_N - 1$, how to make the case $\alpha = 1$ rigorous, or the full shape of P_N for heavy-tailed runs.

3 Aging walks

In this section the walk remembers its age. After its first step it switches direction at step $k \geq 2$ with probability c/k, independently of the past, where $0 < c < 2$. So the walk switches less and less often, and the expected number of switches in N steps is $c(H_N - 1)$, which is about $c \log N$. With a fair first step, A_N/N converges in law to Beta(c, c) [23, Example 1.1 and Theorem 1.4], [24, Theorem 1]. (The chain of [23] is this walk when $c \leq 1$.) The law Beta(c, c) is uniform exactly when $c = 1$, and that case is [24, Remark 2]. At finite N the first step matters. We first give one exact law behind all the ties, then fix the first step, pass to d states, and return to the fair first step.

We write $\mathbb{P}_+$ for the law given $X_1 = +1$, as in Section 1.2, and X_N for the last step. A path is a sequence $x \in \{\pm 1\}^N$, and it switches at time $t \geq 2$ if $x_t \neq x_{t-1}$. Step $k \geq 2$ contributes the factor c/k to the probability of a path if the path switches there, and $1 - c/k$ otherwise. So relative to no switch, a switch at time t multiplies the probability of a path by the odds

$$\omega_t = \frac{c/t}{1 - c/t} = \frac{c}{t - c} \qquad (t \geq 2),$$

and

$$\mathbb{P}_+(X = x) = \prod_{k=2}^{N} \Big(1 - \frac{c}{k}\Big) \prod_{t:\, x_t \neq x_{t-1}} \omega_t \qquad (x_1 = +1). \tag{1}$$

Each ω_t is positive and continuous on $(0, 2)$, increases strictly in c, and tends to 0 as $c \downarrow 0$.

3.1 Drawing afresh or repeating

All the exact laws of this section come from one chain, which works for every weight. Fix $d \geq 2$ and $\theta > 0$, and let $Y_1, Y_2, \ldots$ be the following chain on $\{1, \ldots, d\}$. At step $t \geq 1$, with probability

$$f_t = \frac{d\theta}{d\theta + t - 1}$$

the state Y_t is drawn afresh, uniformly from the d states, and otherwise $Y_t = Y_{t-1}$ (so Y_1 is uniform, since $f_1 = 1$). In other words, at step $t \geq 2$ the chain moves to each other state with probability $\theta/(d\theta + t - 1)$. This is the mechanism of the conservative random walk of Engländer and Volkov [30], who redraw a direction in $\mathbb{Z}^d$ with probability p_n and study recurrence and scaling limits. Let M_T be the vector of visit counts of $Y_1, \ldots, Y_T$, and write $x^{(m)} = x(x + 1) \cdots (x + m - 1)$.

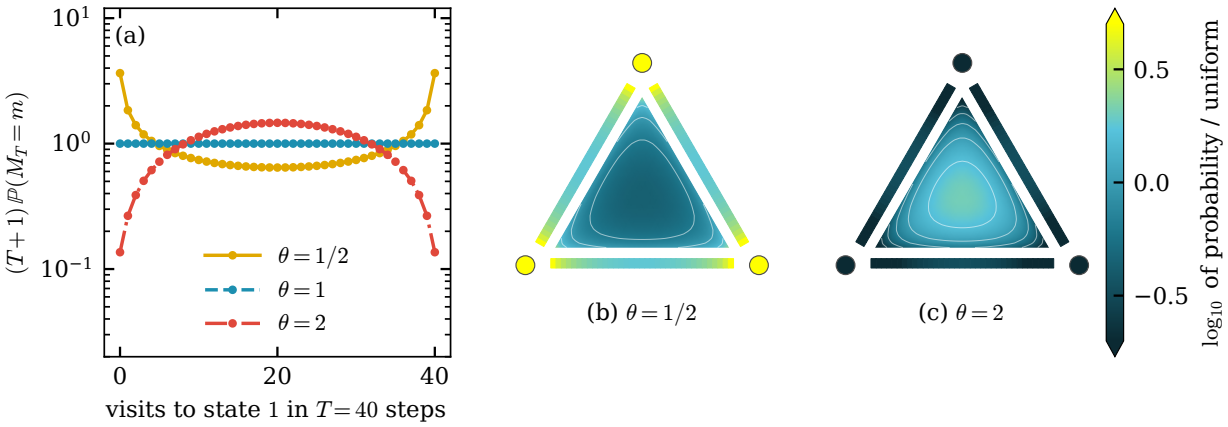


Figure 2: The exact law (2) of the counts of the chain that draws afresh or repeats. (a) Two states and $T = 40$, as a multiple of the uniform law. The law is flat at $\theta = 1$, the ends are the most likely bins for $\theta = \frac{1}{2}$, and the center is for $\theta = 2$. (b), (c) Three states and $T = 36$ on the simplex of visit fractions, colored by the probability of a bin relative to the uniform law. The triangle holds the bins that use all three states, the bands hold the bins that use two, and the disks are the bins that use one. The thin lines are level lines. For $\theta = \frac{1}{2}$ a corner is 9.6 times as likely as under the uniform law and the center 0.43 times. For $\theta = 2$ the numbers are 0.035 and 2.1.

Proposition 3.1 (fresh or repeat)**.** *For every $T \geq 1$, every composition m of T into d nonnegative parts and every state j,*

$$\mathbb{P}(M_T = m,\, Y_T = j) = \frac{m_j}{T}\, Q_T(m), \qquad Q_T(m) = \frac{T!}{(d\theta)^{(T)}} \prod_{i=1}^{d} \frac{\theta^{(m_i)}}{m_i!}. \tag{2}$$

So M_T has the law Q_T of the counts in Pólya's urn with d colors and initial weight θ on each color. For $d = 2$ and even T,

$$\frac{\mathbb{P}(M_T = (T/2, T/2))}{\mathbb{P}(M_T = (T, 0))} = \binom{T}{T/2} \prod_{i=0}^{T/2-1} \frac{\theta + i}{\theta + T/2 + i}, \tag{3}$$

which increases strictly in θ and equals 1 exactly at $\theta = 1$.

In words, the chain repeats its last state where the urn repeats a uniformly chosen earlier draw, and at each fixed time this makes no difference to the counts or to the last state. For two states and even T the center and the end tie at $\theta = 1$, with the ends ahead below it and the center ahead above it, and now the whole law is explicit on both sides of the tie (Figure 2).

Proof. We use induction on T. For $T = 1$ both sides of (2) are $\frac{1}{d}$ when $m = e_j$ and 0 otherwise. Let $T \geq 2$ and suppose (2) holds for $T - 1$. If $m_j = 0$ both sides vanish. Otherwise put $m' = m - e_j$. Summing (2) over the last state gives $\mathbb{P}(M_{T-1} = m') = Q_{T-1}(m')$. A fresh draw gives j with probability $\frac{1}{d}$ whatever the last state was, and a repeat gives j only when the last state was j (this is why the induction carries the last state along with the counts). So

$$\mathbb{P}(M_T = m,\, Y_T = j) = \frac{f_T}{d}\, Q_{T-1}(m') + (1 - f_T)\, \frac{m_j - 1}{T - 1}\, Q_{T-1}(m') = \frac{\theta + m_j - 1}{d\theta + T - 1}\, Q_{T-1}(m').$$

On the other hand $Q_T(m) = Q_{T-1}(m') \cdot \frac{T}{m_j} \cdot \frac{\theta+m_j-1}{d\theta+T-1}$ by the definition of Q_T. This gives (2). Summing over j gives $\mathbb{P}(M_T = m) = Q_T(m)$, and the ratio (3) is $Q_T(T/2, T/2)/Q_T(T, 0)$ with $\theta^{(T)} = \theta^{(T/2)}(\theta + T/2)^{(T/2)}$. Each factor $(\theta + i)/(\theta + T/2 + i)$ increases strictly in θ, and at $\theta = 1$ the product is $(T/2)!^2/T!$. □

The law of the counts alone is known. Dietz, Lippitt and Sethuraman [27, Theorem 2.18] show that the moments of the limit law of the chains of [23] are occupation probabilities at finite N of a second chain, and in the case of a Dirichlet limit that second chain is Y, with the uniform law in the role of their μ and $d\theta$ in the role of their θ. The limit is then the Dirichlet law with every parameter θ [23, Theorem 1.4], and its moments give Q_T. The footnote of [32] quoted below also announces a general connection with Pólya urns. Engländer and Volkov work it out in Chapter 7 of their book [31]. For two states and an integer weight θ they prove (2), with the last state, for every T [31, Theorem 7.2], and they credit this result to E. Crane. What the proposition adds is the joint law with the last state for $d \geq 3$ states, and for a weight θ that is not an integer. The induction needs the last state, and it gives the ties that follow. As $T \to \infty$, (2) turns into the joint limit of the fractions of time and the current state found by Bouguet and Cloez [26, Proposition 4.1], in which the Dirichlet law is size-biased by the coordinate of the current state.

Remark 3.2. Here is another way to see the law of the counts in Proposition 3.1 (but not the joint law with the last state). The fresh draws cut $Y_1, \ldots, Y_T$ into runs, and each run gets a uniform state. The urn can be run in the same way, as a Chinese restaurant process in which each table gets a uniform color. There a table is opened with the same probabilities f_t. The lengths of the runs are the spacings between the successes of independent trials with success probabilities f_t, and it is a classical fact (the Feller coupling, see [25]) that the multiset of these spacings has the same law as the multiset of the table sizes (the two ordered lists do not). This route is different from the one of [31, Chapter 7], which, like Proposition 3.1, goes through the last state. We checked (2) in exact arithmetic for four values of θ, with $T \leq 7$ for $d = 2, 3$ and $T \leq 5$ for $d = 4$ (`verify_aging_urn.py`).

3.2 A fixed first step

With the first step fixed, every interior bin ties with the end exactly at $c = 1$.

Proposition 3.3 (two states, fixed first step)**.** *Let $N \geq 2$ and let the first step be $+1$.*

(a) At $c = 1$, for $1 \leq s \leq N$,

$$\mathbb{P}_+(A_N = s,\, X_N = +1) = \frac{s-1}{N(N-1)}, \qquad \mathbb{P}_+(A_N = s,\, X_N = -1) = \frac{N-s}{N(N-1)}.$$

So A_N is uniform on $\{1, \ldots, N\}$.

(b) For $1 \leq s \leq N-1$ the ratio $\mathbb{P}_+(A_N = s)/\mathbb{P}_+(A_N = N)$ increases strictly in $c \in (0, 2)$, and it equals 1 exactly at $c = 1$.

In words, with a fixed first step every bin from 1 to $N-1$ ties with the end bin N at $c = 1$, for every N. For $c < 1$ the end bin is the most likely bin, and for $c > 1$ it is the least likely of the bins $1, \ldots, N$ (bin 0 has probability 0). So the fixed-start crossing is exactly $c = 1$, and it does not depend on N or on the bin.

Proof. (a) Put $Y_t = X_{t+1}$. Given $X_1 = +1$, step 2 switches with probability $\frac{1}{2}$, so X_2 is uniform, and step $t+1$ switches with probability $1/(t+1)$, which is $\theta/(d\theta+t-1)$ for $d=2$ and $\theta=1$. So Y is the chain of Proposition 3.1 with $d=2$ and $\theta=1$, and there $Q_T(m) = T!/(T+1)! = 1/(T+1)$ for every m. If $A_N = s$, then the counts of $Y_1, \dots, Y_{N-1}$ are $(s-1, N-s)$. So (2) with $T = N-1$ gives $\mathbb{P}_+(A_N = s,\, X_N = +1) = \frac{s-1}{(N-1)N}$ and $\mathbb{P}_+(A_N = s,\, X_N = -1) = \frac{N-s}{(N-1)N}$. Summing over X_N gives $\mathbb{P}_+(A_N = s) = \frac{1}{N}$.

(b) We write the ratio as a sum over paths. Only the path with every step $+1$ has $A_N = N$, and it never switches. So $\mathbb{P}_+(A_N = N)$ is the first product in (1), and dividing by it leaves

$$\frac{\mathbb{P}_+(A_N = s)}{\mathbb{P}_+(A_N = N)} = \sum_{x} \prod_{t:\, x_t \neq x_{t-1}} \omega_t, \tag{4}$$

where the sum is over the paths x with $x_1 = +1$ and s steps equal to $+1$. For example, for $N=3$ and $s=2$ the paths are $(+,+,-)$ and $(+,-,+)$, and the ratio is $\omega_3 + \omega_2\omega_3$. For $1 \le s \le N-1$ there is such a path, and each of them switches at least once. So the right side is a sum of products of the ω_t with at least one factor each. Each ω_t is positive and increases strictly in c, so the sum does too. By (a) it equals 1 at $c=1$. □

After a shift of one step, part (a) is the exercise in [32, Section 3.1, footnote 1]. That footnote also says that the fact can be related to Pólya urns (the authors credit M. Balázs) and that a more general connection of their model with Pólya urns will appear in a later paper. That connection is Chapter 7 of the book [31], and after the same shift part (a), with the last state, is [31, Claim 7.1]. So part (a) is not new. What we add for two states is part (b), and the count law also follows from [27, Theorem 2.18] (Section 3.1). The proof of (b) above is complete in itself. It is the same path-weight argument that gives the unique crossing of the elephant walk in [4], but it does not use that paper. Both parts of Proposition 3.3 are also proved in Lean (Section 4).

3.3 d states

The same thing happens for a chain on d states that leaves its current state for each other state at the same aging rate. For such chains Dietz and Sethuraman [23, Theorem 1.4] proved that the fractions of time converge in law to the Dirichlet law with every parameter θ, which is uniform on the simplex exactly at $\theta = 1$. Bouguet and Cloez [26, Propositions 4.1 and 4.5] find the joint limit of the fractions and the current state. The chain of [23] moves to each other state with probability θ/n at time n once $n > \lceil (d-1)\theta \rceil$, and it does not move before that. Ours, defined below for $\theta < d/(d-1)$, moves to each other state with probability $\theta/(k+d-2)$ at every step $k \ge 2$. We index our chain by their time, so that our step k happens at time $n = k+d-2$ and our chain is in state 1 at time $d-1$. Then the two chains have the same transition probabilities at every time $n \ge d$ with $n > \lceil (d-1)\theta \rceil$, and in particular at every $n \ge d+1$, since $(d-1)\theta < d$. They also have the same limit law of the fractions of time. To see this, put $n_1 = \max(d+1, \lceil d\theta \rceil)$. At every time $n \ge n_1$ each chain stays with probability $1 - d\theta/n$ and otherwise draws a state uniformly from all d states (the draw may give the old state), and here $d\theta/n \le 1$. We couple the two chains from time n_1 on, with the same decisions to draw and the same draws. Since $\sum_n d\theta/n = \infty$, almost surely there is a draw at some time $\tau \ge n_1$, and from time τ on the two chains agree. So up to time $N+d-2$ their vectors of visit counts differ by at most 2τ in total, and their fractions of time differ by $O((\tau+d)/N)$, which tends to 0 almost surely. Hence our fractions of time converge in law to the Dirichlet limit of [23, Theorems 1.3 and 1.4]. At $\theta = 1$, part (a) of the theorem below is an exact finite-N form of their limit. Fix $d \ge 2$ and θ with $0 < \theta < d/(d-1)$. The chain $X_1, X_2, \dots$

on $\{1,\dots,d\}$ starts at $X_1 = 1$. At step $k \geq 2$ it moves to each of the other $d-1$ states with probability $\theta/(k+d-2)$, and it stays with probability $1-(d-1)\theta/(k+d-2)$. The condition on θ says that the probability of staying is positive at every step $k \geq 2$. Let $L_N = (L_N(1),\dots,L_N(d))$ be the vector of visit counts, namely $L_N(i) = \#\{k \leq N : X_k = i\}$. Its values are the compositions $n = (n_1,\dots,n_d)$ of N into d nonnegative parts with $n_1 \geq 1$. There are $\binom{N+d-2}{d-1}$ of them (subtract one from n_1). We call them bins. The corner Ne_1, where e_i is the i-th unit vector, is the bin of the chain that never moves.

Theorem 3.4 (d states)**.** *Let $d \geq 2$, $0 < \theta < d/(d-1)$ and $N \geq 2$.*

(a) At $\theta = 1$, for every bin n and every state j,

$$\mathbb{P}(L_N = n,\, X_N = j) = \frac{n_j - \mathbf{1}\{j=1\}}{(N-1)\binom{N+d-2}{d-1}}. \tag{5}$$

So L_N is uniform on the $\binom{N+d-2}{d-1}$ bins, the corner included.

(b) For every bin $n \neq Ne_1$ the ratio $\mathbb{P}(L_N = n)/\mathbb{P}(L_N = Ne_1)$ increases strictly in θ, and it equals 1 exactly at $\theta = 1$.

In words, at $\theta = 1$ all bins are equally likely, and given the counts the last state is distributed as the state at a uniformly chosen one of the steps $2,\dots,N$ (the forced first visit to state 1 is left out). Every other bin ties with the corner exactly at $\theta = 1$, for every $N \geq 2$ and every d. Below $\theta = 1$ the corner is the most likely bin, and above it the corner is the least likely.

Idea of the proof. At $\theta = 1$ step N stays with weight $N-1$ and moves to each other state with weight 1. We read the weight $N-1$ as $(N-2)+1$. So the chain copies its last state with weight $N-2$, and it picks each of the d states with weight 1. Shifted by one step, this is the chain of Proposition 3.1 with $\theta = 1$, and its counts are uniform. Part (b) is the path-weight argument of Proposition 3.3(b).

Proof. (a) Put $Y_t = X_{t+1}$. At $\theta = 1$ step 2 moves to each other state with probability $\frac{1}{d}$ and stays with probability $1 - \frac{d-1}{d} = \frac{1}{d}$, so X_2 is uniform, and step $t+1$ moves to each other state with probability $1/(t+d-1)$, which is $\theta/(d\theta+t-1)$ at $\theta = 1$. So Y is the chain of Proposition 3.1 with $\theta = 1$, and $M_{N-1} = L_N - e_1$. At $\theta = 1$ we have $Q_T(m) = T!\,(d-1)!/(T+d-1)! = 1/\binom{T+d-1}{d-1}$ for every composition m of T. Every bin n has $n_1 \geq 1$, so $m = n - e_1$ is a composition of $N-1$, and (2) with $T = N-1$ gives (5), since $m_j = n_j - \mathbf{1}\{j=1\}$. Summing over j and using $\sum_j (n_j - \mathbf{1}\{j=1\}) = N-1$ gives $\mathbb{P}(L_N = n) = 1/\binom{N+d-2}{d-1}$.

(b) A path $x \in \{1,\dots,d\}^N$ with $x_1 = 1$ has probability $\prod_{k=2}^{N}\bigl(1 - \frac{(d-1)\theta}{k+d-2}\bigr)$ times the product of the odds

$$\omega_t^{(d)} = \frac{\theta/(t+d-2)}{1-(d-1)\theta/(t+d-2)} = \frac{\theta}{t+d-2-(d-1)\theta}$$

over the times $t \geq 2$ with $x_t \neq x_{t-1}$. The denominator is positive and decreases in θ, so each $\omega_t^{(d)}$ increases strictly in θ. Only the constant path has counts Ne_1. So $\mathbb{P}(L_N = n)/\mathbb{P}(L_N = Ne_1)$ is the sum of the products of the odds over the paths with counts n. For a bin $n \neq Ne_1$ there is such a path, and each of them moves at least once. Hence the ratio increases strictly in θ, and by (a) it equals 1 at $\theta = 1$. □

Remark 3.5. For $d = 2$ the chain switches at step k with probability θ/k, so Theorem 3.4 with $\theta = c$ is Proposition 3.3, and (5) reads $(s-1)/(N(N-1))$ for $j = 1$ and $n = (s, N-s)$.

Take Pólya's urn with d colors, started with one ball of each color. At each draw we take a ball uniformly at random and return it with one more ball of its color. A sequence of $N-1$ draws with counts m has probability $(d-1)!\prod_i m_i!/(N+d-2)!$. So the counts are uniform on the compositions of $N-1$ into d parts, and given the counts m the last draw has color j with probability $m_j/(N-1)$. Hence (5) says that, for each fixed N, the pair $(L_N - e_1, X_N)$ has the law of the counts and the last color after $N-1$ draws. The two processes are different, though. The chain is a Markov chain, and the draws are exchangeable. For $d = 2$ and $N = 4$ the path $1, 1, 2, 2$ has probability $\frac{1}{8}$ for the chain, and the draws $1, 2, 2$ have probability $\frac{1}{12}$. For two states Engländer and Volkov call this relation mimicking [31, Definition 7.1 and Lemma 7.1], and they show with an example of three steps that the coin-turning walk and the urn have different path laws [31, Example 7.1]. Also, at $\theta = 1$ the chain moves at step k with probability $(d-1)/(k+d-2)$, independently over k. This is the probability that customer k opens a new table in the Chinese restaurant process with parameter $d-1$. So the expected number of moves in N steps at the tie is $\sum_{k=2}^{N} \frac{d-1}{k+d-2} = (d-1)\log N + O(1)$, which has the same order, $\log N$, as the number of turns at the crossing from the corners to the center for d directions and runs of geometric length [3].

We checked (5), and the strict ordering in (b) along the five values $\theta = \frac{1}{2}, \frac{9}{10}, 1, \frac{11}{10}, \frac{5}{4}$, in exact rational arithmetic for $d = 2, 3, 4$ and $2 \le N \le 8$ (`verify_aging.py`).

3.4 A fair first step

We return to two states and let the first step be fair. Then both ends can be reached, and $E_N = \mathbb{P}(A_N = N) = \frac{1}{2}\prod_{k=2}^{N}(1 - c/k)$. Changing the sign of every step preserves the law of the walk and sends A_N to $N - A_N$. So

$$\mathbb{P}(A_N = s) = \tfrac{1}{2}\big(\mathbb{P}_+(A_N = s) + \mathbb{P}_+(A_N = N - s)\big), \qquad E_N = \tfrac{1}{2}\mathbb{P}_+(A_N = N), \tag{6}$$

since $\mathbb{P}_+(A_N = 0) = 0$.

Proposition 3.6 (fair first step)**.** *Let $N \ge 2$ and $1 \le s \le N-1$, with a fair first step.*

(a) At $c = 1$, $\mathbb{P}(A_N = s) = \frac{1}{N}$ and $E_N = \frac{1}{2N}$.

(b) The ratio $\mathbb{P}(A_N = s)/E_N$ is continuous and strictly increasing in $c \in (0, 2)$. It tends to 0 as $c \downarrow 0$ and equals 2 at $c = 1$. So there is exactly one c in $(0, 2)$ with $\mathbb{P}(A_N = s) = E_N$, and it is less than 1.

For even N we write c_N for the crossing in (b) with $s = h = N/2$, so that $C_N = E_N$ exactly at $c = c_N$. In words, at $c = 1$ every interior bin still has probability $\frac{1}{N}$, and the two ends share the probability $\frac{1}{N}$ that the fixed start gave to one end. So at $c = 1$ the center is twice as likely as an end, and the crossing moves below 1. Part (a) is the case $a = 0$ of the exact laws that Engländer and Volkov give for the switch probabilities $1/(n+a)$ [31, Section 5.7].

Proof. (a) This follows from (6) and Proposition 3.3(a). (b) By (6) the ratio is the sum of the right sides of (4) for s and for $N - s$. Both are finite sums of products of at least one ω_t. So the ratio is continuous and strictly increasing, and it tends to 0 as $c \downarrow 0$. By (a) it equals 2 at $c = 1$. Hence it equals 1 exactly once, at some $c < 1$. □

How far below 1 is c_N? The end is of order N^{-c} and the center is of order $1/N$, so they meet when N^{1-c} is about 2.

Proposition 3.7 (the fair-start crossing)**.** *For even $N \to \infty$,*

$$c_N = 1 - \frac{\log 2}{\log N + \gamma + 2 - 2\log 2} + O\big((\log N)^{-3}\big),$$

where γ is Euler's constant.

Idea of the proof. A monotonicity argument first traps c_N within $O(1/\log N)$ of 1, so an expansion at $c = 1$ applies. The end is explicit, namely $E_N = \Gamma(N+1-c)/(2\Gamma(2-c)\Gamma(N+1))$, which is $N^{-c}/(2\Gamma(2-c))$ with relative error $O(1/N)$. For the center we prove a local limit theorem with a rate, namely $NC_N = d_c + O(N^{-0.4})$ uniformly for $c \in [c^{(3)}, 1]$, where $c^{(3)} = (9-\sqrt{57})/2 \approx 0.725$ and $d_c = 4^{1-c}/\mathrm{B}(c,c)$ is the density of $\mathrm{Beta}(c,c)$ at $\frac{1}{2}$. So $\log(C_N/E_N) = (c-1)\log N + \Theta(c) + O(N^{-0.4})$ with $\Theta(c) = \log(2\Gamma(2-c)d_c)$. Here $\Theta(1) = \log 2$ and $\Theta'(1) = \gamma + 2 - 2\log 2$, and solving for the zero gives the expansion. The proof is in Appendix B. The local limit theorem comes from an averaging identity for C_N and a coupling of the walk with a sign process driven by a Poisson process of intensity $c\,dt/t$, whose occupation fraction is exactly $\mathrm{Beta}(c,c)$.

The error term decays only like $(\log N)^{-3}$, but the expansion is already accurate at moderate N. The computed crossing at $N = 200$ is $c_{200} = 0.8925$, against 0.8932 from the expansion without its error term. The constant in the error term is not computed.

4 Verification

All computations are in Python. The values of R_N in Table 1 were computed in two ways, both from Proposition 2.2, namely by its formula and by a spectral method that reads each factor off as a Taylor coefficient of a generating function. For small N they were also checked against a recursion on the first run and a sum over all run compositions, which do not use Proposition 2.2. At $N = 10^5$ the formula is used only in the full run of the scripts, which takes about twenty minutes, and the quick run uses the spectral method alone there. The same holds for the check that R_N increases at every even N up to 10^5 at $\alpha = \varphi$, which the quick run makes up to 10^4. The aging probabilities, and with them the crossing c_{200}, were computed in three ways, namely by a forward recursion, from transfer matrices at roots of unity, and from the sums over paths in (4). In exact rational arithmetic $C_{200} - E_{200}$ is negative at $c_{200} - 10^{-6}$ and positive at $c_{200} + 10^{-6}$. No proof in this paper rests on a computation. The tables, the figures and the numbers quoted outside the proofs rest on computation alone. The script `verify_heavy_runs.py` recomputes the numbers of Section 2 and Appendix A, the script `verify_aging.py` those of Section 3 and Appendix B, and `verify_aging_urn.py` checks Proposition 3.1 in exact arithmetic for two, three and four states. The numbers in the caption of Figure 2 come from `figures/make_aging_figure.py`.

Many statements of this paper are also proved in Lean 4.33.1 with Mathlib at commit `0df444a360ea`, in the folder `kagey-problems/problem131/lean/PaperD`, which is shared with an earlier version of this work. The file `MAP.md` there lists each statement of this paper by its label, with its Lean names, its status in Lean and the source of each classical input. The kernel checks every proof, and every theorem depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. Table 3 lists the results that Lean proves. Some of them rest on the six classical results of Table 4, which Lean states as named hypotheses, and nothing else is assumed.

Two proofs in Lean differ from ours. In Step 4 of the proof of Theorem 2.9, Lean finds $\int_0^\infty tg(t)^2\,dt$ from $\mathbb{E}G^{i\tau}$ by Parseval's identity for the density of $\log G$, so it does not identify the law of $G_1/(G_1+G_2)$. For Proposition B.5 Lean uses Proposition 3.1 with $d = 2$ and $\theta = c$ in place of the coupling, so it does not need Proposition B.3 or Lemma B.4. Lean does not formalize the

Table 3: The statements proved in Lean, with the abbreviations of Table 4.

Results	Status in Lean	Notes
Section 2		
Propositions 2.1 and 2.2	proved	For every law of D, and with $N \geq 1$ in Proposition 2.1
Lemma 2.3	(i) from POLY, (ii) from POLY, LEVY and FI, (iii) from GN, (iv) proved	Part (ii) without the Laplace transform of G and the remark on its skewness and scale, and part (iv) for the value of I_α
Theorem 2.4	from GN, POLY, LEVY and FI	For every first run D
Corollary 2.5	from GN, POLY, LEVY and FI	With $\Gamma(\varphi) \geq 0.8856$ in place of the minimum of Γ, and without the decimals of $C(\varphi)$ and $1/C(\varphi)$. Lean also proves $C(\varphi) < 0.952$
Theorem 2.9	from GN, POLY, LAP, FI and LERCH	Step 4 by a different route (see below)
Section 3		
Propositions 3.1, 3.3 and 3.6, Theorem 3.4	proved	Proposition 3.1 for every d and every $\theta > 0$
Remark 3.5	proved	The exact statements and the bound $(d-1)\log N + O(1)$
Proposition 3.7	proved	Also the term of order L^{-3} at the end of Appendix B, but not the numbers at $N = 200$
Appendices		
Lemma A.1	proved for $1 < \alpha < 2$	The case that Theorem 2.4 uses
Lemma A.2, Proposition A.3	proved for $a_m = m^\kappa$	The case that Theorem 2.4 uses
Lemma A.4	(i) from POLY, LAP and FI, (ii) from GN, (iii) and (iv) proved	
Lemma B.1, Propositions B.2 and B.5	proved	Proposition B.5 by a different proof (see below), with the rate $\sqrt{\log N/N}$

Table 4: The classical results that Lean states as hypotheses.

Abbr.	Result	Source
GN	local limit theorem of Gnedenko	[17, Theorem 4.2.1]
POLY	expansion of $\mathrm{Li}_s(e^\mu)$ near $\mu = 0$	[22, 25.12.12]
LEVY	continuity theorem of Lévy	[20, Section XV.3]
FI	Fourier inversion for an integrable characteristic function	[20, Section XV.3]
LAP	continuity theorem for Laplace transforms	[20, Section XIII.1]
LERCH	uniqueness theorem of Lerch for Laplace transforms	[41, Chapter II]

six classical results, Theorem 2.8, Proposition B.3, Lemma B.4, Remarks 2.6, 2.7, 3.2 and 2.10, or the numbers that rest on computation.

The scripts and the Lean files are in the repository named under "Code availability" at the end of the paper, which also says how to rerun the checks.

Acknowledgments. I thank Peter Kagey for posing the problem and Aidan Deshong for insightful conversations.

A note on the use of AI

All mathematical claims, proofs and computations in this paper are my responsibility.

A Proofs for Section 2

We use the notation of Section 2, with $\tau_0 = 0$. For a first run D let $\tau_m^D = D + \tau'_{m-1}$, where (τ'_m) is an independent copy of (τ_m), and $\tau_0^D = 0$. Then $u_m^D(k) = \mathbb{P}(\tau_m^D = k)$ and $q_m^D(k) = \mathbb{P}(\tau_{m-1}^D < k \le \tau_m^D)$. We find each of these by conditioning on where the run before ends. For example, the m-th run covers level k exactly when the first $m-1$ runs end at some level $k-j$ with $j \ge 1$ and the m-th run has length at least j. In this way, for $m, k \ge 1$,

$$q_m(k) = \sum_{j \ge 1} \bar{F}(j)\, u_{m-1}(k-j), \qquad q_{m+1}^D(k) = \sum_{j \ge 1} \bar{F}(j)\, u_m^D(k-j),$$
$$u_m^D(k) = \sum_{\ell \ge 1} \mathbb{P}(D = \ell)\, u_{m-1}(k-\ell). \tag{7}$$

Here $u_0(k) = \mathbf{1}\{k = 0\}$. The sums τ_m and τ_m^D increase strictly in m. So exactly one run covers a given level $k \ge 1$, and at most one run ends there. Hence $\sum_m q_m(k) = \sum_m q_m^D(k) = 1$ and $\sum_m u_m(k) \le 1$, $\sum_m u_m^D(k) \le 1$. Also, the first m' runs reach level k exactly when one of them covers it, so $\sum_{m \le m'} q_m(k) = \mathbb{P}(\tau_{m'} \ge k)$, and in the same way $\sum_{m \le m'} q_m^D(k) = \mathbb{P}(\tau_{m'}^D \ge k)$.

Sections A.1 to A.4 treat $\alpha > 1$ and prove Theorems 2.4 and 2.8 together. Section A.5 treats $0 < \alpha < 1$ and proves Theorem 2.9.

A.1 The setting for $\alpha > 1$

Let $\alpha > 1$, so that $\mu = \zeta(\alpha) < \infty$. The proof of the center asymptotics uses the exponent only through the following three facts about a norming sequence (a_m) of positive numbers.

(F1) For $m \ge 3$ we have $a_m = a(m)$, where $a(x) = c_a x^\kappa (\log x)^\beta$ on $[3, \infty)$ with constants $c_a > 0$, $0 < \kappa < 1$ and $\beta \ge 0$.

(F2) The variables $(\tau_m - m\mu)/a_m$ converge in law to a variable G that has a bounded, uniformly continuous density g with $g(z) \to 0$ as $|z| \to \infty$. With $z_{m,k} = (k - m\mu)/a_m$, the numbers $\delta_m = \sup_k |a_m u_m(k) - g(z_{m,k})|$ tend to 0, and $\bar{u} = \sup_{m \ge 1} \sup_k a_m u_m(k)$ is finite.

(F3) There is a constant C_B such that $u_m(k) \le C_B\, m\, k^{-\alpha-1}$ for all $m \ge 1$ and $k \ge 2m\mu$.

The function a in (F1) is increasing, and $a(\lambda x)/a(x) \to \lambda^\kappa$ as $x \to \infty$, uniformly for λ in a compact subset of $(0, \infty)$. So $a_m \to \infty$ and $a_m/a_{m-1} \to 1$. For $1 < \alpha < 2$ we take $a_m = m^{1/\alpha}$. Then (F1)

holds with $\kappa = 1/\alpha$ and $\beta = 0$, (F2) is Lemma 2.3(ii) and (iii), and (F3) is Lemma A.1 below. We write ω for the modulus of continuity of g and $\bar{g}(x) = \sup_{|y|\geq x} g(y)$. So $\omega(\eta) \to 0$ as $\eta \to 0$ and $\bar{g}(x) \to 0$ as $x \to \infty$.

A.2 Two lemmas on renewal sums

A sum of m runs that is much larger than its mean is most likely made by a single long run. The first lemma turns this into a local bound. Bounds of this kind are classical in the big-jump domain [19]. We include a short proof to keep the paper self-contained.

Lemma A.1. *Let $\alpha > 1$. There is a constant $C_B = C_B(\alpha)$ such that $u_m(k) \leq C_B\, m\, k^{-\alpha-1}$ for all $m \geq 1$ and $k \geq 2m\mu$.*

Proof. We split the event $\{\tau_m = k\}$ according to whether some run is long, namely at least $y = k/r$, where $r > 2$ is a constant that depends only on α and is chosen below. The first part is small because one run takes a given value $\ell \geq y$ with probability at most $\alpha y^{-\alpha-1}$. If no run is long, then τ_m is a sum of variables below y whose mean is at most $k/2$, and a Chernoff bound shows that such a sum rarely reaches k. Here C depends only on α and may change from line to line. Since $\mu \geq 1$ we have $m \leq k/2$.

One big jump. By the mean value theorem $\mathbb{P}(\xi = \ell) = \ell^{-\alpha} - (\ell+1)^{-\alpha} \leq \alpha\ell^{-\alpha-1}$. Fix an index i. Then

$$\mathbb{P}(\tau_m = k,\, \xi_i \geq y) = \sum_{\ell \geq y} \mathbb{P}(\xi = \ell)\, \mathbb{P}(\tau_m - \xi_i = k - \ell) \leq \alpha y^{-\alpha-1} = \alpha r^{\alpha+1} k^{-\alpha-1},$$

because the first factor is at most $\alpha y^{-\alpha-1}$ and the second factors add up to at most 1. The m indices together give at most $\alpha r^{\alpha+1} m k^{-\alpha-1}$.

No big jump. We may take $y \geq 1$, since otherwise $\max_i \xi_i < y$ is impossible. Let $x_0 > 0$ and $s \geq 0$. For $0 \leq x \leq s$ we have $e^x \leq 1 + x + \frac{x^2}{2} e^s$. We use this with $x = s\xi/y$ on the event $\{\xi < y\}$ and then $1 + x \leq e^x$. This gives $\mathbb{E}[e^{s\xi/y}; \xi < y] \leq \exp\bigl(\frac{s}{y}\mathbb{E}[\xi; \xi < y] + \frac{s^2 e^s}{2y^2}\mathbb{E}[\xi^2; \xi < y]\bigr)$. By independence and Markov's inequality for $e^{s\tau_m/y}$,

$$\mathbb{P}\bigl(\tau_m \geq x_0,\ \max_{i \leq m} \xi_i < y\bigr) \leq \exp\Bigl(-\frac{s x_0}{y} + m\Bigl(\frac{s}{y}\,\mathbb{E}[\xi; \xi < y] + \frac{s^2 e^s}{2y^2}\,\mathbb{E}[\xi^2; \xi < y]\Bigr)\Bigr). \tag{8}$$

We take $x_0 = k$. Since $\mathbb{E}[\xi; \xi < y] \leq \mu$ and $k - m\mu \geq k/2$, the first two terms of the exponent add up to at most $-\frac{s}{y} \cdot \frac{k}{2} = -sr/2$. So the exponent is at most $-sr/2 + \frac{1}{2} r^2 s^2 e^s\, m\, k^{-2}\, \mathbb{E}[\xi^2; \xi < y]$. It remains to choose s and r so that the bound is at most $Cmk^{-\alpha-1}$. The choice depends on how fast the truncated second moment grows, so there are three cases.

Let $1 < \alpha < 2$. Then $\mathbb{E}[\xi^2; \xi < y] \leq \alpha \sum_{\ell<y} \ell^{1-\alpha} \leq Cy^{2-\alpha}$, so the exponent is at most $-sr/2 + Cr^\alpha s^2 e^s m/k^\alpha$. Take $s = \frac{1}{2}\log(k^\alpha/m)$, which is positive since $m \leq k/2 < k^\alpha$. Then $e^s m/k^\alpha = e^{-s}$ and $s^2 e^{-s} \leq 1$, so the second term is at most Cr^α. The first term is $-\frac{r}{4}\log(k^\alpha/m)$. So the probability is at most $C(m/k^\alpha)^{r/4}$. We take $r = 4 + 4/(\alpha - 1)$, so that $r/4 = 1 + 1/(\alpha - 1)$. Since $m \leq k$, the bound is at most $C(m/k^\alpha)(k^{1-\alpha})^{1/(\alpha-1)} = Cmk^{-\alpha-1}$.

Let $\alpha > 2$. Then $\mathbb{E}[\xi^2; \xi < y] \leq \mathbb{E}\xi^2 < \infty$, so the exponent is at most $-sr/2 + Cr^2 s^2 e^s m/k^2$. Take $s = \frac{1}{2}\log(k^2/m)$, which is positive since $m < k^2$. As before the probability is at most $C(m/k^2)^{r/4}$. With $r = 4\alpha$ and $m \leq k$ this is $C\, m\, m^{\alpha-1} k^{-2\alpha} \leq Cmk^{-\alpha-1}$.

Let $\alpha = 2$. Then $\mathbb{E}[\xi^2; \xi < y] \leq 2\sum_{\ell<y} \ell^{-1} \leq 2\log(ek)$, so the exponent is at most $-sr/2 + r^2 s^2 e^s m \log(ek)/k^2$. Take $s = \frac{1}{2}\log\frac{k^2}{m\log(ek)}$, which is positive since $m\log(ek) \leq \frac{1}{2}k\log(ek) < k^2$.

As before the probability is at most $C(m\log(ek)/k^2)^{r/4}$. With $r=20$ and $m\le k$ this is at most $Cmk^{-6}\log^5(ek)\le Cmk^{-3}$.

In each case the two bounds together give the lemma. □

Lemma A.2. *Let $\alpha>1$ and assume* (F1) *and* (F2). *Let D be any a.s. finite first run. Then*

$$\sup_k\Big|\frac{a_m}{\mu}q_m(k)-g(z_{m,k})\Big|,\qquad \sup_k\big|a_m u_m^D(k)-g(z_{m,k})\big|,\qquad \sup_k\Big|\frac{a_m}{\mu}q_{m+1}^D(k)-g(z_{m,k})\Big|$$

tend to 0 as $m\to\infty$. For $m\ge 2$, $\sup_k q_m(k)\le \mu\bar u/a_{m-1}$ and $\sup_k u_m^D(k)\le \bar u/a_{m-1}$.

Proof. By (7), q_m and u_m^D are sums of shifts of u_{m-1} with total weights $\sum_j \bar F(j)=\mu$ and 1. Since $u_{m-1}(k)\le \bar u/a_{m-1}$, this gives the two bounds for $m\ge 2$.

For the limits we use that a shift by a bounded amount j moves the point $z_{m,k}$ only by $O(1/a_m)$, and that g is uniformly continuous. So the local limit for u_{m-1} passes to these sums. We give the details for q_m. Fix $J\ge 1$ and $Z>0$, put $\sigma_J=\sum_{j>J}\bar F(j)$ and write $z=z_{m,k}$. We compare term by term the two sums

$$a_{m-1}q_m(k)=\sum_{j\ge1}\bar F(j)\,a_{m-1}u_{m-1}(k-j)\qquad\text{and}\qquad \mu g(z)=\sum_{j\ge1}\bar F(j)\,g(z).$$

The terms with $j>J$ add up to at most $\bar u\sigma_J$ in the first sum and $\|g\|_\infty\sigma_J$ in the second.

Let $j\le J$. By (F2), $|a_{m-1}u_{m-1}(k-j)-g(z_{m-1,k-j})|\le\delta_{m-1}$. Next we compare $g(z_{m-1,k-j})$ with $g(z)$. The definition of $z_{m,k}$ gives $z_{m-1,k-j}=z\,a_m/a_{m-1}+(\mu-j)/a_{m-1}$. If $|z|\le Z$, then $|z_{m-1,k-j}-z|\le\eta_m$ with $\eta_m=Z|a_m/a_{m-1}-1|+(J+\mu)/a_{m-1}$, so the two values of g differ by at most $\omega(\eta_m)$. If $|z|>Z$, then $|z_{m-1,k-j}|\ge Z/2$ once m is large (how large depends only on J and Z), so both values lie in $[0,\bar g(Z/2)]$. In both cases they differ by at most $\omega(\eta_m)+\bar g(Z/2)$. Hence, uniformly in k for large m,

$$\big|a_{m-1}q_m(k)-\mu g(z)\big|\le\mu\big(\delta_{m-1}+\omega(\eta_m)+\bar g(Z/2)\big)+(\bar u+\|g\|_\infty)\sigma_J.$$

Here $\eta_m\to0$ as $m\to\infty$. So we let $m\to\infty$ and then $J,Z\to\infty$, and the right side tends to 0. Since $a_m/a_{m-1}\to1$ and $a_{m-1}q_m(k)\le\mu\bar u$, we may replace a_{m-1} by a_m on the left. This gives the first limit.

The second limit follows in the same way from the sum for u_m^D, with the weights $\mathbb{P}(D=\ell)$ and with $\mathbb{P}(D>J)$ in place of σ_J, which tends to 0 as $J\to\infty$ because D is a.s. finite. The third follows from the sum for q_{m+1}^D, with $a_m u_m^D$ in place of $a_{m-1}u_{m-1}$, the second limit in place of $\delta_{m-1}\to0$, and the shift $z_{m,k-j}-z_{m,k}=-j/a_m$. For the terms with $j>J$ we use the bound $a_m u_m^D(k)\le\bar u\,a_m/a_{m-1}$ from above in place of $\bar u$, and a_m/a_{m-1} is bounded since it tends to 1. □

A.3 The center for $\alpha>1$

Proposition A.3. *Let $\alpha>1$ and assume* (F1), (F2) *and* (F3). *Let D be any a.s. finite first run and put $b_h=a(h/\mu)$. As $N=2h\to\infty$,*

$$C_N\sim\frac{2}{b_h}\int_{\mathbb{R}}g(z)^2\,dz.$$

Idea of the proof. We split the sum in Proposition 2.2 by the number m of runs into four regions. In the main region $|m\mu - h| \le Mb_h$ both factors are given by the local limit theorem, and the terms form a Riemann sum for $2b_h^{-1}\int_{-M}^{M} g^2$. For $m \le h/(4\mu)$ the level h is far above the mean of τ_m, and (F3) makes these terms $O(h^{-\alpha})$. In the other two regions (the m between these two, and the m above the main region) one factor of each term is $O(1/b_h)$, and the other factors add up to a tail probability of G, which is small when M is large.

Proof. By Proposition 2.2, $C_N = \sum_{m\ge1}\big[u_m(h)q^D_{m+1}(h) + u^D_m(h)q_m(h)\big]$. Fix $M > 1$, put $m_\pm = (h \pm Mb_h)/\mu$, and let $\Sigma_1, \dots, \Sigma_4$ be the parts of this sum over the regions $\mathcal{R}_1 = \{m \le h/(4\mu)\}$, $\mathcal{R}_2 = \{h/(4\mu) < m < m_-\}$, $\mathcal{R}_3 = \{m_- \le m \le m_+\}$ and $\mathcal{R}_4 = \{m > m_+\}$. We show that $b_h\Sigma_3$ tends to $2\int_{-M}^{M} g^2$ and that the other three parts are small.

We need three consequences of (F1). First, $b_h = o(h)$ and $h^{-\alpha} = o(1/b_h)$, since $\kappa < 1 < \alpha$. Second, $a(h/(8\mu)) \ge b_h/9$ and $a(h/(2\mu)) \ge b_h/3$ for large h, since $8^{-\kappa} > \frac19$ and $2^{-\kappa} > \frac13$. Third, $a_m/b_h \to 1$ uniformly in m with $|m\mu - h| \le 2Mb_h$.

Region $\mathcal{R}_1$ (far too few runs). Here $h \ge 4m\mu$, so level h is far above the mean of τ_m. We show that $u_m(h)$ and $q_m(h)$ are both $O(h^{-\alpha})$ for $m \in \mathcal{R}_1$. By (F3), $u_m(h) \le C_B m h^{-\alpha-1} \le C_B h^{-\alpha}/(4\mu)$. Also $q_1(h) = h^{-\alpha}$. For $m \ge 2$ we split the sum for $q_m(h)$ in (7) at $j = h/2$. The terms with $j \ge h/2$ have $\bar F(j) \le (h/2)^{-\alpha}$, and the factors $u_{m-1}(h-j)$ add up to at most 1. So these terms add up to at most $(h/2)^{-\alpha}$. For $j < h/2$ we have $h - j > h/2 \ge 2\mu(m-1)$, so (F3) applies to $u_{m-1}(h-j)$, and the weights $\bar F(j)$ add up to at most μ. So the other terms add up to at most $\mu C_B(m-1)(h/2)^{-\alpha-1} \le 2^{\alpha-1}C_B h^{-\alpha}$. In Σ_1 each of these factors is multiplied by $q^D_{m+1}(h)$ or by $u^D_m(h)$, and $\sum_m q^D_{m+1}(h) \le 1$ and $\sum_m u^D_m(h) \le 1$. Hence $\Sigma_1 = O(h^{-\alpha})$, which is $o(1/b_h)$.

Region $\mathcal{R}_2$ (too few runs). In each term one factor is at most $9\bar u/b_h$, and the other factors add up to the probability that few runs reach level h. For large h we have $m - 1 \ge h/(8\mu)$ here, so $a_m \ge a_{m-1} \ge b_h/9$ and $u_m(h), u^D_m(h) \le 9\bar u/b_h$ by (F2) and Lemma A.2. Let $m' = \lfloor m_-\rfloor$. Every $m \in \mathcal{R}_2$ is at most m', so $\sum_{m\in\mathcal{R}_2} q_m(h) \le \mathbb{P}(\tau_{m'} \ge h)$ and $\sum_{m\in\mathcal{R}_2} q^D_{m+1}(h) \le \mathbb{P}(D + \tau'_{m'} \ge h)$. The last probability is at most $\mathbb{P}(D \ge b_h) + \mathbb{P}(\tau_{m'} \ge h - b_h)$. Since $m'\mu \le h - Mb_h$ and $a_{m'} \le b_h$, the event $\{\tau_{m'} \ge h - b_h\}$ implies $(\tau_{m'} - m'\mu)/a_{m'} \ge M - 1$. Now $\mathbb{P}(D \ge b_h) \to 0$ because D is a.s. finite, and $(\tau_{m'} - m'\mu)/a_{m'}$ converges in law to G, which has a continuous law. So $\limsup_h b_h\Sigma_2 \le 18\bar u\,\mathbb{P}(G \ge M - 1)$.

Region $\mathcal{R}_4$ (too many runs). This is like $\mathcal{R}_2$. For large h we have $m - 1 \ge h/(2\mu)$ here, so $u_m(h), u^D_m(h) \le 3\bar u/b_h$. Let $m'' = \lfloor m_+\rfloor$. The runs after the first m'' cover level h exactly when the first m'' do not reach it. So $\sum_{m\in\mathcal{R}_4} q_m(h) = \mathbb{P}(\tau_{m''} < h)$ and, since $D \ge 1$, $\sum_{m\in\mathcal{R}_4} q^D_{m+1}(h) = \mathbb{P}(D + \tau'_{m''} < h) \le \mathbb{P}(\tau_{m''} < h)$. Now $m''\mu > h + Mb_h - \mu$, so $(h - m''\mu)/a_{m''} \le (\mu - Mb_h)/a_{m''} \to -M$. Hence $\limsup_h b_h\Sigma_4 \le 6\bar u\,\mathbb{P}(G \le -M)$.

Region $\mathcal{R}_3$ (the main part). Here $m \to \infty$ and $a_m = b_h(1 + o(1))$ uniformly. By (F2) and Lemma A.2, the factors $u_m(h)$ and $u^D_m(h)$ are $(g(z_{m,h}) + o(1))/a_m$, and the factors $q_m(h)$ and $q^D_{m+1}(h)$ are $\mu(g(z_{m,h}) + o(1))/a_m$. So, uniformly in $m \in \mathcal{R}_3$,

$$u_m(h)q^D_{m+1}(h) + u^D_m(h)q_m(h) = \frac{2\mu}{a_m^2}\big(g(z_{m,h})^2 + o(1)\big).$$

Put $y_m = (h - m\mu)/b_h$. Then $z_{m,h} = y_m b_h/a_m$ and $|y_m| \le M$, so $|z_{m,h} - y_m| \le M|b_h/a_m - 1| \to 0$ and $g(z_{m,h})^2 = g(y_m)^2 + o(1)$. The y_m with $m \in \mathcal{R}_3$ are the points of an arithmetic progression with step μ/b_h that lie in $[-M, M]$. There are at most $2Mb_h/\mu + 1$ of them, so the $o(1)$ terms below add up to $o(1)$, and the rest is a Riemann sum. Hence

$$b_h\Sigma_3 = 2\big(1 + o(1)\big)\frac{\mu}{b_h}\sum_{m\in\mathcal{R}_3}\big(g(y_m)^2 + o(1)\big) \to 2\int_{-M}^{M} g^2.$$

Adding the four regions,

$$\limsup_{h\to\infty}\Big|b_hC_N - 2\int g^2\Big| \le 2\int_{|z|>M} g^2 + 18\bar{u}\,\mathbb{P}(G \ge M-1) + 6\bar{u}\,\mathbb{P}(G \le -M),$$

and letting $M \to \infty$ gives $C_N \sim 2b_h^{-1}\int g^2$. □

Proof of Theorem 2.4. Let $1 < \alpha < 2$ and $a_m = m^{1/\alpha}$. As noted in Section A.1, (F1) to (F3) hold, with $b_h = (h/\mu)^{1/\alpha}$. So Proposition A.3 gives $C_N \sim 2I_\alpha(\mu/h)^{1/\alpha}$, and the second form follows from Lemma 2.3(iv) and $h = N/2$. □

A.4 Proof of Theorem 2.8

We apply Proposition A.3 with the standard normal density in place of the stable density. So let $\alpha \ge 2$ and let g be the standard normal density. We have to find a norming sequence (a_m) with (F1), (F2) and (F3). Fact (F3) is Lemma A.1, which holds for every $\alpha > 1$. For (F2) it is enough that $(\tau_m - m\mu)/a_m$ converges in law to the standard normal law. Indeed, g is bounded and uniformly continuous and vanishes at $\pm\infty$. Every $\mathbb{P}(\xi = k)$ is positive, so ξ has maximal span 1, and Gnedenko's local limit theorem [17, Theorem 4.2.1] turns the convergence in law into $\delta_m \to 0$. Then $\bar{u} < \infty$ follows as in the proof of Lemma 2.3(iii). So in each of the two cases we choose (a_m), check (F1) and the normal limit, and read off $b_h = a(h/\mu)$. Since $\int g^2 = 1/(2\sqrt{\pi})$, Proposition A.3 then gives $C_N \sim 1/(\sqrt{\pi}\,b_h)$.

(a) Let $\alpha > 2$. Then ξ has a finite variance, and we use the central limit theorem. Summation by parts gives $\mathbb{E}\xi^2 = \sum_{k\ge1}(2k-1)\bar{F}(k) = 2\zeta(\alpha-1) - \zeta(\alpha)$, so $v = \mathbb{E}\xi^2 - \mu^2$ is as stated and finite. It is positive because ξ is not constant. We take $a_m = \sqrt{vm}$. This has (F1) with $\kappa = \frac12$ and $\beta = 0$, and the central limit theorem gives the normal limit. So $b_h = \sqrt{vh/\mu}$, and with $h = N/2$ we get $C_N \sim (\pi vh/\mu)^{-1/2} = \sqrt{2\mu/(\pi vN)}$. For the stationary start Proposition 2.1 gives $E_N = N^{1-\alpha}(1 + O(1/N))/(2(\alpha-1)\mu)$. Dividing E_N by C_N gives $R_N \sim C_2(\alpha)N^{3/2-\alpha}$.

(b) Let $\alpha = 2$, so that $\mu = \zeta(2)$ and $\mathbb{E}\xi^2 = \infty$. The central limit theorem with the norming $\sqrt{m}$ fails, but the variance is infinite only by a logarithm, and the limit is still normal with a slightly larger norming. To see this we compute the truncated second moment

$$U(x) = \mathbb{E}[\xi^2; \xi \le x] = \sum_{k\le x} k^2\big(k^{-2} - (k+1)^{-2}\big) = \sum_{k\le x}\frac{2k+1}{(k+1)^2} = 2\log x + O(1).$$

This function varies slowly. So ξ is in the domain of attraction of the normal law, and $(\tau_m - m\mu)/a_m$ converges in law to the standard normal law for every sequence (a_m) with $mU(a_m)/a_m^2 \to 1$ [20, Section XVII.5]. We try $a(x) = \sqrt{x\log x}$. Then $U(a(m)) = \log m + \log\log m + O(1)$ and $a(m)^2/m = \log m$, so $mU(a(m))/a(m)^2 \to 1$. We take $a_m = a(m)$ for $m \ge 3$ and $a_1 = a_2 = 1$. This has (F1) with $\kappa = \beta = \frac12$. So $b_h^2 = (h/\mu)\log(h/\mu) \sim N\log N/(2\mu)$ and $C_N \sim \sqrt{2\mu/(\pi N\log N)}$. For the stationary start $E_N = N^{-1}(1 + O(1/N))/(2\mu)$ by Proposition 2.1. Dividing gives $R_N \sim \sqrt{\pi\log N/(8\mu^3 N)}$.

Finally we take the fresh start. In both cases it has the end $E_N = \frac12 N^{-\alpha}$ by Proposition 2.1, and the center asymptotics are the same because Proposition A.3 holds for every a.s. finite first run. So R_N is of order $N^{1/2-\alpha}$ for $\alpha > 2$ and of order $N^{-3/2}(\log N)^{1/2}$ for $\alpha = 2$, and it tends to 0. □

A.5 Proof of Theorem 2.9

Let $0 < \alpha < 1$ with a fresh start, $a_m = m^{1/\alpha}$ and $z_{m,k} = k/a_m$. There is no centering now. In this section μ is not used, and constants may depend on α.

Lemma A.4. *Let $0 < \alpha < 1$.*

(i) τ_m/a_m converges in law to a positive variable G with $\mathbb{E}e^{-sG} = e^{-\Gamma(1-\alpha)s^\alpha}$ for $s \ge 0$. The law of G has a bounded, uniformly continuous density g that vanishes on $(-\infty, 0]$ and at $+\infty$.

(ii) (Gnedenko) The constant $\bar{u} = \sup_{m\ge1} \sup_k a_m u_m(k)$ is finite, and $\delta_m = \sup_k |a_m u_m(k) - g(k/a_m)| \to 0$ as $m \to \infty$.

(iii) There is a constant C_B such that $u_m(k) \le C_B\, m\, k^{-\alpha-1}$ for all $m \ge 1$ and $k \ge a_m$.

(iv) $g(y) \le C_B y^{-1-\alpha}$ for $y > 1$. So $\int_0^\infty t\, g(t)^2\, dt < \infty$.

Proof. (i) Summation by parts gives $1 - \mathbb{E}x^\xi = (1-x)\sum_{k\ge1} \bar{F}(k)x^{k-1}$ for $0 \le x < 1$. With $x = e^{-v}$ this is $1 - \mathbb{E}e^{-v\xi} = (e^v - 1)\mathrm{Li}_\alpha(e^{-v})$, and [22, 25.12.12] gives $\mathrm{Li}_\alpha(e^{-v}) = \Gamma(1-\alpha)v^{\alpha-1} + \zeta(\alpha) + O(v)$ as $v \downarrow 0$. So $1 - \mathbb{E}e^{-v\xi} = \Gamma(1-\alpha)v^\alpha + O(v)$. As $1/\alpha > 1$, this gives $m(1 - \mathbb{E}e^{-s\xi/a_m}) \to \Gamma(1-\alpha)s^\alpha$ for each $s \ge 0$. Hence $\mathbb{E}e^{-s\tau_m/a_m} \to e^{-\Gamma(1-\alpha)s^\alpha}$, and the continuity theorem for Laplace transforms [20, Section XIII.1] gives the limit $G \ge 0$. Both sides of $\mathbb{E}e^{-sG} = e^{-\Gamma(1-\alpha)s^\alpha}$ are continuous on $\mathrm{Re}\, s \ge 0$ and analytic on $\mathrm{Re}\, s > 0$, with the principal branch on the right. So the identity holds at $s = -i\theta$, and $|\mathbb{E}e^{i\theta G}| = \exp\bigl(-\Gamma(1-\alpha)\cos(\frac{\pi\alpha}{2})|\theta|^\alpha\bigr)$ is integrable. Fourier inversion and the Riemann–Lebesgue lemma give g. It vanishes on $(-\infty, 0)$ because $G \ge 0$, and at 0 by continuity.

(ii) As in Lemma 2.3(iii), ξ has maximal span 1, and Gnedenko's local limit theorem [17, Theorem 4.2.1] gives $\delta_m \to 0$. Then $a_m u_m(k) \le \|g\|_\infty + 1$ for large m and $a_m u_m(k) \le a_m$ for the others.

(iii) We argue as in Lemma A.1, with $r = 4 + 4/\alpha$, $y = k/r$ and $t = k^\alpha/m$, which is at least 1 because $k \ge a_m$. For infinite mean, local bounds of this kind are in Doney [18]. One thing is new. For k of order a_m the claim is $u_m(k) \le C/a_m$, which is a local bound, and a bound for the tail of τ_m cannot give it. So we cut the m runs into two halves. One half must be large, which we bound as before. The other half must then take one exact value, and this gives the factor $1/a_m$.

For $m = 1$ the claim is $\mathbb{P}(\xi = k) \le \alpha k^{-\alpha-1}$. Let $m \ge 2$. The bound $\alpha r^{\alpha+1} m k^{-\alpha-1}$ for the event that some run is at least y holds as before. For the other event we may take $y \ge 1$. Split the indices into $I_1 = \{1, \dots, \lfloor m/2 \rfloor\}$ and $I_2 = \{\lfloor m/2 \rfloor + 1, \dots, m\}$, and write $\tau_I = \sum_{i\in I} \xi_i$. On $\{\tau_m = k\}$ one of τ_{I_1}, τ_{I_2} is at least $k/2$, and then the other one equals k minus it. So by independence

$$\mathbb{P}\bigl(\tau_m = k,\ \max_i \xi_i < y\bigr) \le \sum_{i=1,2} \mathbb{P}\bigl(\tau_{I_i} \ge k/2,\ \max_{j\in I_i} \xi_j < y\bigr)\ \sup_\ell \mathbb{P}(\tau_{I_{3-i}} = \ell).$$

Each half has at least $\lfloor m/2 \rfloor \ge m/3$ indices, so by (ii) the supremum is at most $\bar{u}/a_{\lfloor m/2 \rfloor} \le 3^{1/\alpha}\bar{u}/a_m$. For the first factor we use (8) for the sum over I_i, with $x_0 = k/2$ and at most m terms. Now $\mathbb{E}[\xi; \xi < y] \le \sum_{\ell<y} \bar{F}(\ell) \le Cy^{1-\alpha}$ and $\mathbb{E}[\xi^2; \xi < y] \le Cy^{2-\alpha}$, and $my^{-\alpha} = r^\alpha/t$. So the exponent in (8) is at most $-sr/2 + C(s + s^2e^s)/t$. We take $s = \frac{1}{2}\log t$. Then $e^s = \sqrt{t}$, so the second term is bounded, and the first is $-\frac{r}{4}\log t$. Hence the first factor is at most $Ct^{-r/4} = Ct^{-1-1/\alpha}$. Since $a_m = m^{1/\alpha}$, we have $t^{-1-1/\alpha}/a_m = mk^{-\alpha-1}$, which is the claim.

(iv) Fix $y > 1$ and let $k_m = \lceil y a_m \rceil$. By (iii) and $ma_m^{-\alpha} = 1$, $a_m u_m(k_m) \le C_B m a_m (ya_m)^{-\alpha-1} = C_B y^{-1-\alpha}$. By (ii) and the continuity of g, the left side tends to $g(y)$ as $m \to \infty$. The integral converges because g is bounded and $tg(t)^2 \le C_B^2 t^{-1-2\alpha}$ for $t > 1$. □

Idea of the proof of Theorem 2.9. With a fresh start, Proposition 2.2 reads $C_N = \sum_{m\ge1} u_m(h)\big[q_m(h)+q_{m+1}(h)\big]$. Put $t_m = h/a_m$. Step 1 finds $q_m(h)$ on the scale where t_m is of order 1, namely $mq_m(h) \approx t_m g(t_m)/\alpha$. Step 2 sums over m. The terms with t_m very large or very small are negligible, and the others form a Riemann sum in the variable $\log m$. This gives $NC_N \to 4\int_0^\infty tg(t)^2\,dt$. Step 3 identifies this limit as the density at $\frac12$ of $Y = G_1/(G_1+G_2)$, where G_1, G_2 are independent copies of G. Step 4 finds the law of Y from the moments of G_1/G_2, and with it the value of the limit.

Proof of Theorem 2.9. Fix $M > 1$.

Step 1. Let $g_1(t) = \int_0^t w^{-\alpha} g(t-w)\,dw$. We claim that $mq_m(h) = g_1(t_m) + o(1)$ as $h \to \infty$, uniformly in m with $t_m \in [1/(2M), 2M]$, and that $g_1(t) = tg(t)/\alpha$. The reason for the first claim is that the sum for q_m in (7) is the convolution of the tail $j^{-\alpha}$ with u_{m-1}, and on the scale a_m it becomes the convolution g_1. The m in this range tend to infinity with h. Since $|k/a_{m-1} - k/a_m| \le 2M(a_m/a_{m-1}-1) \to 0$ for $0 \le k \le h$, Lemma A.4(ii) for u_{m-1} and the uniform continuity of g give $a_m u_{m-1}(k) = g(k/a_m) + o(1)$ uniformly in $0 \le k \le h$. As $m\,j^{-\alpha} = (j/a_m)^{-\alpha}$ and $u_{m-1}(0) = 0$, the sum for q_m in (7) becomes

$$mq_m(h) = \frac{1}{a_m}\sum_{j=1}^{h-1}\Big(\frac{j}{a_m}\Big)^{-\alpha}\Big(g\big(t_m - \tfrac{j}{a_m}\big) + o(1)\Big).$$

For $x > 0$ we have $\frac{1}{a_m}\sum_{j\le xa_m}(j/a_m)^{-\alpha} \le \int_0^x w^{-\alpha}dw = x^{1-\alpha}/(1-\alpha)$. With $x = t_m \le 2M$ this shows that the $o(1)$ terms add up to $o(1)$. With $x = \eta$ it shows that the terms with $j \le \eta a_m$ are $O(\eta^{1-\alpha})$, and so is $\int_0^\eta w^{-\alpha} g(t_m - w)\,dw$. The other terms form a Riemann sum with mesh $1/a_m$ of the function $w^{-\alpha}g(t-w)$, which is uniformly continuous on $\{\eta \le w \le t \le 2M\}$. Letting $\eta \to 0$ proves the first claim.

For the second claim we compare Laplace transforms. The transform of the convolution g_1 is the product of those of $w^{-\alpha}$ and g, namely $\Gamma(1-\alpha)s^{\alpha-1}e^{-\Gamma(1-\alpha)s^\alpha}$. This is $\frac{1}{\alpha}$ times $-\frac{d}{ds}e^{-\Gamma(1-\alpha)s^\alpha}$, and minus the derivative of the transform of g is the transform of $tg(t)$. Both g_1 and $tg(t)$ are continuous on $(0,\infty)$, so $g_1(t) = tg(t)/\alpha$.

Step 2. We sum $u_m(h)[q_m(h)+q_{m+1}(h)]$ over m in three ranges. If $t_m > M$, then $h > Ma_m$, so $m < (h/M)^\alpha$, and Lemma A.4(iii) gives $u_m(h) \le C_B m h^{-\alpha-1} \le C_B M^{-\alpha}/h$. Since $\sum_m (q_m(h)+q_{m+1}(h)) \le 2$, these m contribute at most $2C_BM^{-\alpha}/h$ to C_N. If $t_m < 1/M$, then $u_m(h) \le \bar u/a_m < \bar u/(Mh)$, and these m contribute at most $2\bar u/(Mh)$. For the remaining m we have $t_{m+1} = t_m(1+o(1)) \in [1/(2M), 2M]$ and $(m+1)/m \to 1$, and g_1 is uniformly continuous on $[1/(2M), 2M]$. So Step 1 applies to $q_m(h)$ and to $q_{m+1}(h)$, and both are $(t_m g(t_m)/\alpha + o(1))/m$. With $1/a_m = t_m/h$ and $\delta_m \to 0$ we get, uniformly,

$$u_m(h)\big[q_m(h)+q_{m+1}(h)\big] = \frac{t_m}{h}\big(g(t_m)+o(1)\big)\,\frac{2}{m}\Big(\frac{t_m g(t_m)}{\alpha} + o(1)\Big) = \frac{2}{\alpha hm}\big(t_m^2 g(t_m)^2 + o(1)\big).$$

As m runs over these values, $\log m = \alpha\log(h/t_m)$ runs over an interval of length $2\alpha\log M$ with steps $\log(1+1/m) = (1+o(1))/m$. So the sum of $1/m$ is $O(\log M)$, the $o(1)$ terms contribute $o(1/h)$, and the rest is a Riemann sum in the variable $\log m$. With $dm/m = -\alpha\,dt/t$ it is $\frac{2}{\alpha h}\big(\int_{1/M}^M t^2 g(t)^2\,\frac{\alpha\,dt}{t} + o(1)\big)$. Hence

$$hC_N = 2\int_{1/M}^M tg(t)^2\,dt + O(M^{-\alpha} + M^{-1}) + o(1),$$

where the O term is uniform in h. By Lemma A.4(iv) we may let $M \to \infty$, and we get $NC_N = 2hC_N \to 4\int_0^\infty tg(t)^2\,dt$.

Step 3. Let G_1, G_2 be independent copies of G and $Y = G_1/(G_1+G_2)$. The map $(G_1,G_2) \mapsto (Y, G_1+G_2)$ has inverse $(y,r) \mapsto (yr,(1-y)r)$ with Jacobian r. So Y has the density $f(y) = \int_0^\infty r\,g(yr)\,g((1-y)r)\,dr$ on $(0,1)$. With $r = 2t$ we get $f(\frac12) = \int_0^\infty r\,g(r/2)^2\,dr = 4\int_0^\infty tg(t)^2\,dt$. Hence $NC_N \to f(\frac12)$ by Step 2. The function f is continuous on $[\frac14, \frac34]$. Indeed, for such y both yr and $(1-y)r$ are at least $r/4$, so by Lemma A.4(iv) the integrand is at most $r\min(\|g\|_\infty^2, Cr^{-2-2\alpha})$. This is integrable, and dominated convergence applies.

Step 4. It remains to show that $f(\frac12) = \ell_\alpha(\frac12)$. We show that Y has the density ℓ_α. To do this we compute the moments $\mathbb{E}W^p$ of $W = G_1/G_2$ for small $|p|$, and we check that a certain explicit density ρ_α has the same moments. The law of Y does not change if G is multiplied by a constant, so we may take $\mathbb{E}e^{-sG} = e^{-s^\alpha}$. For $x > 0$ and $0 < p < 1$,

$$x^p = \frac{p}{\Gamma(1-p)}\int_0^\infty s^{-p-1}(1-e^{-sx})\,ds, \qquad x^{-p} = \frac{1}{\Gamma(p)}\int_0^\infty s^{p-1}e^{-sx}\,ds.$$

We put $x = G$, take expectations and substitute $w = s^\alpha$. The first formula gives $\mathbb{E}G^p$ for $0 < p < \alpha$ and the second gives $\mathbb{E}G^{-p}$ for $0 < p < 1$. Together, $\mathbb{E}G^p = \Gamma(1-p/\alpha)/\Gamma(1-p)$ for $|p| < \alpha$ (see also [21, Theorem 2.6.3]). Now $Y = W/(1+W)$ and $\mathbb{E}W^p = \mathbb{E}G^p\,\mathbb{E}G^{-p}$ by independence. With $\Gamma(1-z)\Gamma(1+z) = \pi z/\sin(\pi z)$, which is the reflection formula [22, 5.5.3] combined with $\Gamma(1+z) = z\Gamma(z)$, this is

$$\mathbb{E}W^p = \frac{\Gamma(1-\frac{p}{\alpha})\Gamma(1+\frac{p}{\alpha})}{\Gamma(1-p)\Gamma(1+p)} = \frac{\sin\pi p}{\alpha\sin(\pi p/\alpha)}, \qquad 0 < |p| < \alpha.$$

Let $\rho_\alpha(w) = \frac{\sin\pi\alpha}{\pi}\,\frac{w^{\alpha-1}}{w^{2\alpha}+2w^\alpha\cos\pi\alpha+1}$ for $w > 0$. We use the classical integral

$$\int_0^\infty \frac{x^\beta\,dx}{x^2+2x\cos\vartheta+1} = \frac{\pi\sin\beta\vartheta}{\sin\vartheta\,\sin\pi\beta}, \qquad 0 < |\beta| < 1,\ 0 < \vartheta < \pi,$$

whose value at $\beta = 0$ is the limit $\vartheta/\sin\vartheta$. For $-1 < \beta < 0$ it follows from the partial fractions $\frac{1}{x^2+2x\cos\vartheta+1} = \frac{1}{2i\sin\vartheta}\big(\frac{1}{x+e^{-i\vartheta}} - \frac{1}{x+e^{i\vartheta}}\big)$ and $\int_0^\infty x^\beta(x+e^{\pm i\vartheta})^{-1}dx = -\pi e^{\pm i\beta\vartheta}/\sin(\pi\beta)$. The substitution $x \mapsto 1/x$ shows that the left side is even in β, and so is the right side. So the formula holds for $0 < \beta < 1$ too. We use it with $x = w^\alpha$, $\beta = p/\alpha$ and $\vartheta = \pi\alpha$. Then $w^{\alpha-1}dw = dx/\alpha$, and we get

$$\int_0^\infty w^p\rho_\alpha(w)\,dw = \frac{\sin\pi\alpha}{\pi\alpha}\cdot\frac{\pi\sin\pi p}{\sin\pi\alpha\,\sin(\pi p/\alpha)} = \mathbb{E}W^p \qquad (0 < |p| < \alpha).$$

At $p = 0$ the value $\vartheta/\sin\vartheta$ of the classical integral gives $\int_0^\infty \rho_\alpha = 1$, so ρ_α is a probability density. So $\log W$ and the logarithm of a variable with density ρ_α have the same moment generating function near 0. A moment generating function that is finite near 0 determines the law, so W has density ρ_α. Then $Y = W/(1+W)$ has density $\rho_\alpha(\frac{y}{1-y})(1-y)^{-2}$, and multiplying the numerator and the denominator by $(1-y)^{2\alpha}$ shows that this is $\ell_\alpha(y)$. Both f and ℓ_α are densities of Y that are continuous near $\frac12$, so

$$f(\tfrac12) = \ell_\alpha(\tfrac12) = \frac{2\sin\pi\alpha}{\pi(1+\cos\pi\alpha)} = \frac{2}{\pi}\tan\frac{\pi\alpha}{2}.$$

Dividing $E_N = \frac12 N^{-\alpha}$ by C_N gives the formula for R_N. □

B Proof of Proposition 3.7

We use the setting of Section 3 with two states and a fair first step. So $0 < c < 2$, X_1 is a fair sign, and the switch indicators $\chi_k = \mathbf{1}\{X_k \neq X_{k-1}\}$, $k \geq 2$, are independent Bernoulli(c/k) variables, independent of X_1. Put

$$\rho(a,b) = \prod_{k=a+1}^{b} \Big(1 - \frac{c}{k}\Big) = \frac{\Gamma(b+1-c)\,\Gamma(a+1)}{\Gamma(b+1)\,\Gamma(a+1-c)}, \qquad \lambda_{t,N} = \frac{c}{t}\,\rho(t,N)$$

for integers $1 \leq a \leq b$ and $2 \leq t \leq N$. So $\lambda_{t,N}$ is the probability of a switch at time t and none in $(t,N]$, and $E_N = \frac{1}{2}\rho(1,N)$. Let T_s^+ be the time of the s-th step equal to $+1$. Let B be a Beta(c,c) variable, $d_c = 4^{1-c}/\mathrm{B}(c,c)$ its density at $\frac{1}{2}$, and $\Theta(c) = \log\big(2\Gamma(2-c)\,d_c\big)$. For even N we write $h = N/2$ and $C_N = \mathbb{P}(A_N = h)$. Changing the sign of every step preserves the law of the walk and sends A_N to $N - A_N$. We call this flip symmetry.

Idea of the proof. A walk that ends in the center with last step -1 takes its h-th step $+1$ at some time r, switches at time $r+1$, and never switches again. So C_N is an average of the explicit numbers $\lambda_{r+1,N}$ over the law of T_h^+ (Lemma B.1). For $c \leq 1$ these numbers are monotone in r, which already traps c_N between two explicit roots and gives $1 - c_N = O(1/\log N)$ (Proposition B.2). For the constant we need the law of T_h^+. We have $T_h^+ \leq r$ exactly when $A_r \geq h$, and A_r/r is close to a Beta(c,c) variable with an error that is a power of r (Lemma B.4). This is proved by coupling the walk with a sign process in continuous time that switches at the points of a Poisson process of intensity $c\,dt/t$, whose occupation fraction is exactly Beta(c,c) at every time (Proposition B.3). Summation by parts then gives $NC_N = d_c + O(N^{-0.4})$ (Proposition B.5).

B.1 A gamma ratio

Gautschi's inequality [22, 5.6.4] says that $x^{1-s} < \Gamma(x+1)/\Gamma(x+s) < (x+1)^{1-s}$ for $x > 0$ and $0 < s < 1$. We claim that for $0 < c < 2$ and real $y \geq 2$,

$$\frac{\Gamma(y+1-c)}{\Gamma(y+1)} = y^{-c}(1+\vartheta) \qquad \text{with} \qquad -\frac{1}{y} \leq \vartheta \leq \frac{2}{y}. \tag{9}$$

If $c = 1$ the left side is $1/y$ and $\vartheta = 0$. If $c < 1$ we take $x = y$ and $s = 1 - c$. Then the left side lies between $(y+1)^{-c}$ and y^{-c}, and $(1+1/y)^{-c} \geq 1 - c/y \geq 1 - 1/y$. If $c > 1$ we take $x = y$ and $s = 2 - c$ and use $\Gamma(y+1-c) = \Gamma(y+2-c)/(y+1-c)$. Then the left side lies between $(y+1)^{1-c}/(y+1-c)$ and $y^{1-c}/(y+1-c)$. Here $y/(y+1-c) = 1 + (c-1)/(y+1-c) \leq 1 + 1/(y-1) \leq 1 + 2/y$, and $(1+1/y)^{1-c}\,y/(y+1-c) \geq (1+1/y)^{-1} \geq 1 - 1/y$.

B.2 Averaging at the center and a bracket

Lemma B.1 (averaging at the center)**.** *Let $N \geq 2$ be even and $T = T_h^+$. Then*

$$C_N = 2\sum_{r=h}^{N-1} \mathbb{P}(T = r)\,\lambda_{r+1,N} \qquad \textit{and} \qquad \sum_{r=h}^{N-1} \mathbb{P}(T = r) = \frac{1}{2}. \tag{10}$$

If $0 < c \leq 1$, then $c/N \leq C_N \leq \lambda_{h+1,N}$.

Proof. On the event $\{A_N = h,\, X_N = -1\}$ the last step equal to $+1$ is the h-th one, at a time $T = r \le N-1$. It is followed by a switch at $r+1$ and no switch in $(r+1, N]$. Conversely, these conditions give $A_N = h$ and $X_N = -1$. The event $\{T = r\}$ depends only on $X_1, \chi_2, \dots, \chi_r$. So $\mathbb{P}(A_N = h,\, X_N = -1) = \sum_{r=h}^{N-1} \mathbb{P}(T = r)\, \lambda_{r+1,N}$, and flip symmetry gives the same value for $X_N = +1$. Next, $T \le N-1$ exactly when $A_{N-1} \ge h$. This has probability $\frac{1}{2}$, because $N-1$ is odd and A_{N-1} is symmetric about $(N-1)/2$. Finally $\lambda_{t+1,N}/\lambda_{t,N} = t/(t+1-c)$, which is at most 1 for $c \le 1$. So $\lambda_{N,N} = c/N \le \lambda_{t,N} \le \lambda_{h+1,N}$ for $h+1 \le t \le N$, and by (10) C_N is a weighted average of these values. □

For an integer $M \ge 2$ put $\eta_M(c) = M\rho(1, M)/(2c)$. Since $\rho(1, M)$ decreases in c and equals $1/M$ at $c = 1$, the function η_M decreases strictly on $(0, 1]$, from ∞ at 0 to $\eta_M(1) = \frac{1}{2}$. Let $c^{(M)} \in (0, 1)$ be the root of $\eta_M = 1$.

Proposition B.2 (bracket)**.** *The numbers $c^{(M)}$ increase with M, and $1 - c^{(M)} = \frac{\log 2}{\log M + 1 + \gamma} + O\big((\log M)^{-3}\big)$. For every even $N \ge 2$ we have $c^{(h+1)} \le c_N \le c^{(N)}$. So $(1 - c_N) \log N \to \log 2$, and $c_N \ge c^{(3)} = (9 - \sqrt{57})/2 > 0.725$ for $N \ge 4$.*

Proof. The equation $\eta_M(c) = 1$ says that $\frac{1}{2}\rho(1, M) = c/M$, namely that the end for M steps equals c/M. We compare E_N with the two bounds for C_N in Lemma B.1, and each comparison is an inequality for some η_M.

Monotonicity. For $c < 1$ we have $\eta_{M+1}/\eta_M = (M + 1 - c)/M > 1$. So $\eta_{M+1}(c^{(M)}) > 1$, and since η_{M+1} decreases, $c^{(M+1)} > c^{(M)}$.

The bracket. By Proposition 3.6(b), $C_N < E_N$ for $c < c_N$ and $C_N > E_N$ for $c > c_N$, and $c_N < 1$. So it is enough to compare C_N with E_N for $0 < c < 1$. Let $c^{(N)} < c < 1$. Then $\eta_N(c) < 1$, that is $E_N = \frac{1}{2}\rho(1, N) < c/N$, and $c/N \le C_N$ by Lemma B.1. So $C_N > E_N$, which means $c > c_N$. Hence $c_N \le c^{(N)}$. Now let $c < c^{(h+1)}$. Then $\eta_{h+1}(c) > 1$, that is $\frac{1}{2}\rho(1, h+1) > c/(h+1)$. With $\rho(1, N) = \rho(1, h+1)\rho(h+1, N)$ this gives $E_N > \frac{c}{h+1}\rho(h+1, N) = \lambda_{h+1,N} \ge C_N$. So $c < c_N$. Hence $c_N \ge c^{(h+1)}$.

The expansion. The number $c^{(3)}$ solves $3\rho(1, 3) = (2 - c)(3 - c)/2 = 2c$, that is $c^2 - 9c + 6 = 0$. For $N \ge 4$ we have $h + 1 \ge 3$, so $c_N \ge c^{(h+1)} \ge c^{(3)}$. By (9) with $y = M$ we have $M\rho(1, M) = M^{1-c}(1 + O(1/M))/\Gamma(2 - c)$. So the equation $\eta_M(c) = 1$ reads $(1 - c)\log M = \Theta_0(c) + O(1/M)$ with $\Theta_0(c) = \log\big(2c\,\Gamma(2 - c)\big)$, uniformly for $0 < c \le 1$. For $M \ge 3$ we have $c^{(M)} \in [c^{(3)}, 1)$, where Θ_0 is bounded. So $\varepsilon = 1 - c^{(M)}$ is $O(1/\log M)$, and we may expand at $c = 1$. We have $\Theta_0(1) = \log 2$ and $\Theta_0'(1) = 1 - \psi(1) = 1 + \gamma$, where $\psi = \Gamma'/\Gamma$ and $\psi(1) = -\gamma$ [22, 5.4.12]. Taylor's theorem gives $\varepsilon(\log M + 1 + \gamma) = \log 2 + O(\varepsilon^2) + O(1/M)$, which is the expansion. The limit of $(1 - c_N)\log N$ follows since $\log(h + 1) \sim \log N$. □

B.3 The continuum occupation

Proposition B.3 (continuum occupation)**.** *Let Π be a Poisson process on $(0, \infty)$ with intensity $c\,dt/t$, and let $Y(1)$ be a fair sign independent of Π. Put $Y(t) = Y(1)(-1)^{\Pi((1,t])}$ for $t \ge 1$ and $Y(t) = Y(1)(-1)^{\Pi((t,1])}$ for $0 < t < 1$. Then for every $r > 0$ the fraction $Z_r = \frac{1}{r}\int_0^r \mathbf{1}\{Y(t) = +1\}\,dt$ has exactly the Beta(c, c) law.*

This agrees with [32, Section 6.1], where a remark obtains the Beta(c, c) one-dimensional marginals of the zigzag process defined there from a scaling limit and mentions an unpublished direct proof by E. Crane. We give a direct proof.

Proof. We pass to logarithmic time, where the sign process is stationary, and then we condition on the last switch before time 1. This gives an equation for the law of Z_1 that has only one solution, and we check that Beta laws solve it.

Step 1 (the law of Z_r does not depend on r). The points $\log t$, $t \in \Pi$, form a Poisson process $\tilde\Pi$ of rate c on $\mathbb{R}$, and $W(u) = Y(e^u)$ satisfies $W(v) = W(u)(-1)^{\tilde\Pi((u,v])}$ for $u < v$. Fix a real ℓ. Then $W(\ell)$ is $W(0)$ times a function of $\tilde\Pi$, and $W(0)$ is a fair sign independent of $\tilde\Pi$. So $W(\ell)$ is a fair sign independent of $\tilde\Pi$ too. As $\tilde\Pi - \ell$ has the law of $\tilde\Pi$, the pair $(W(\ell), \tilde\Pi - \ell)$ has the law of $(W(0), \tilde\Pi)$, and so the process $W(\ell + \cdot)$ has the law of W. With $\ell = \log r$ this says that $Y(r\,\cdot)$ has the law of Y. Hence $Z_r = \int_0^1 \mathbf{1}\{Y(rt) = +1\}\,dt$ has the law of Z_1.

Step 2 (an equation for the law of Z_1). The substitution $t = e^{-s}$ gives $Z_1 = \int_0^\infty e^{-s}\mathbf{1}\{V_s = +1\}\,ds$, where $V_s = W(-s)$. The process V looks backward from time 1. It starts at a fair sign and switches at the points of a Poisson process of rate c. Let $\pi_\pm$ be the law of Z_1 given $V_0 = \pm 1$, let τ be the first switch of V, and let $U = e^{-\tau}$ (in the original time, U is the last switch of Y before time 1). Then $\mathbb{P}(U \le u) = \mathbb{P}(\tau \ge -\log u) = u^c$ for $0 \le u \le 1$, so U is Beta$(c, 1)$. After τ the process V starts afresh from $-V_0$, independently of τ. We split the integral at τ. Given $V_0 = +1$ the part before τ is $\int_0^\tau e^{-s}ds = 1 - U$, and the part after τ is U times a copy of Z_1 for a process started at -1. Given $V_0 = -1$ the part before τ is 0, and the part after is U times a copy of Z_1 for a process started at $+1$. So π_+ is the law of $1 - U + UZ_-$ and π_- is the law of UZ_+, where $Z_\pm$ has law $\pi_\pm$ and is independent of U. In other words, (π_+, π_-) is a fixed point of the map Ψ that sends a pair (ν_+, ν_-) of laws on $[0, 1]$ to the laws of $1 - U + UZ_-$ and UZ_+, where $Z_\pm \sim \nu_\pm$ are independent of U.

Step 3 (Ψ has only one fixed point). We measure the distance between two laws ν, ν' on $[0, 1]$ by the Wasserstein distance, namely the smallest value of $\mathbb{E}|Z - Z'|$ over all pairs with $Z \sim \nu$ and $Z' \sim \nu'$. Take two pairs (ν_+, ν_-) and (ν'_+, ν'_-), and take $Z_\pm \sim \nu_\pm$ and $Z'_\pm \sim \nu'_\pm$ that attain these distances, independently of U. Then $\mathbb{E}|UZ_\pm - UZ'_\pm| = \mathbb{E}U\,\mathbb{E}|Z_\pm - Z'_\pm|$. So Ψ multiplies the larger of the two distances by at most $\mathbb{E}U = c/(c+1) < 1$. Two fixed points are then at distance 0, so they are equal.

Step 4 (Beta laws solve the equation). We claim that the pair Beta$(c+1, c)$, Beta$(c, c+1)$ is a fixed point of Ψ. (This is the natural guess. If the sign is $+$ at time 1, then more of the time before is spent at $+$.) We need one fact, namely that UZ is Beta$(c, c+1)$ whenever Z is Beta$(c+1, c)$ and independent of U. To prove it let G_a, G_b, G_d be independent Gamma variables with shapes a, b, d and unit scale. A change of variables shows that $G_a/(G_a + G_b)$ is Beta(a, b) and independent of $G_a + G_b$. Hence it is independent of the pair $(G_a + G_b, G_d)$ and of $(G_a + G_b)/(G_a + G_b + G_d)$, which is Beta$(a+b, d)$. Their product $G_a/(G_a + G_b + G_d)$ is Beta$(a, b+d)$. With $(a, b, d) = (c, 1, c)$ this is the fact.

Now let Z_+ be Beta$(c+1, c)$ and Z_- be Beta$(c, c+1)$, independent of U. By the fact UZ_+ is Beta$(c, c+1)$. Also $1 - Z_-$ is Beta$(c+1, c)$, so $U(1 - Z_-)$ is Beta$(c, c+1)$ and $1 - U + UZ_- = 1 - U(1 - Z_-)$ is Beta$(c+1, c)$. This proves the claim. By Step 3, π_+ is Beta$(c+1, c)$ and π_- is Beta$(c, c+1)$. Since V_0 is a fair sign, Z_1 has the equal mixture of these two laws. As $\mathrm{B}(c+1, c) = \mathrm{B}(c, c+1) = \frac{1}{2}\mathrm{B}(c, c)$, the mixture has density $x^{c-1}(1-x)^{c-1}(x + 1 - x)/\mathrm{B}(c, c)$, which is the Beta$(c, c)$ density. □

Lemma B.4 (coupling)**.** *Let $J \subset (0, 2)$ be compact, $a_J = \min(\min J, 1)$ and $\kappa_J = a_J/(1 + a_J)$. There is C_J such that for $c \in J$ and every integer $r \ge 2$,*

$$\sup_{y \in \mathbb{R}} \big|\mathbb{P}(A_r \ge y) - \mathbb{P}(rB \ge y)\big| \le C_J\,(\log r)\,r^{-\kappa_J}.$$

Proof. We build the walk and the process Y of Proposition B.3 on one probability space so that $X_k = Y(k)$ for most $k \le r$. This works because Y changes sign between the times $k-1$ and k with probability close to c/k. Then A_r is close to rZ_r, which has the law of rB. The first K steps are left uncoupled, and we choose K at the end.

Step 1 (the coupling). Take Π and Y as in Proposition B.3 and an integer $1 \le K \le r$. For $k > K$ let $\chi'_k = \mathbf{1}\{\Pi((k-1,k]) \text{ is odd}\}$, so that $Y(k) = Y(k-1)(-1)^{\chi'_k}$. These variables depend on Π in disjoint intervals, so they are independent of each other and of $Y(K)$. The number $\Pi((k-1,k])$ is Poisson with mean $\nu = c\log\frac{k}{k-1}$, and a Poisson variable with mean ν is odd with probability $\frac12(1-e^{-2\nu})$. So $\mathbb{P}(\chi'_k = 1) = p'_k = \frac12\bigl(1-(1-1/k)^{2c}\bigr)$. By Taylor's theorem for the function $\frac12(1-(1-x)^{2c})$ at $x = 1/k$, $|p'_k - c/k| = \frac12 c|2c-1|(1-\vartheta)^{2c-2}k^{-2}$ for some $0 < \vartheta < 1/k$. This is at most $3k^{-2}$. Indeed, if $c \ge 1$ then $\frac12 c|2c-1| < 3$ and the power of $1-\vartheta$ is at most 1. If $c < 1$ then $\frac12 c|2c-1| \le \frac12$ and the power is at most $(1-\frac12)^{-2} = 4$, as $k \ge 2$.

For $K < k \le r$ we couple $\chi_k \sim \text{Bernoulli}(c/k)$ with χ'_k so that $\mathbb{P}(\chi_k \ne \chi'_k) = |p'_k - c/k|$, using independent extra randomness. We put $X_K = Y(K)$ and $X_k = X_{k-1}(-1)^{\chi_k}$ for $K < k \le r$, and we draw $X_1, \dots, X_{K-1}$ from the law of the walk given X_K, again with independent randomness. In the walk X_K is a fair sign and $(\chi_k)_{k>K}$ is independent of $(X_1, \dots, X_K)$. The same holds here, so $(X_1, \dots, X_r)$ has the law of the walk.

Step 2 (A_r is close to rZ_r). Let $\mathcal{A}$ be the event that $\chi_k = \chi'_k$ for $K < k \le r$. Then $\mathbb{P}(\mathcal{A}^c) \le \sum_{k>K} 3k^{-2} \le 3/K$, and on $\mathcal{A}$ we have $X_k = Y(k)$ for $K \le k \le r$. Now A_r is the sum of $\mathbf{1}\{X_k = +1\}$ over the steps $k \le r$, and rZ_r is the sum of $\int_{k-1}^k \mathbf{1}\{Y(t) = +1\}\,dt$ over the same k. Let $K < k \le r$. If $\Pi((k-1,k]) = 0$, then $Y = Y(k)$ on $(k-1,k]$, so on $\mathcal{A}$ the two terms for this k are equal. The other $k \le r$ number at most $K + \Pi((K,r])$, and for each of them the two terms differ by at most 1. Hence $|A_r - rZ_r| \le K + \Pi((K,r])$ on $\mathcal{A}$. The variable $\Pi((K,r])$ is Poisson with mean $c\log(r/K) \le 2\log r$, so by Markov's inequality it exceeds K with probability at most $2\log r/K$. So with $b_K = (3 + 2\log r)/K$ we have $|A_r - rZ_r| \le 2K$ outside an event of probability at most b_K.

Step 3 (the distribution functions). By Step 2, for every real y,

$$\mathbb{P}(rZ_r \ge y + 2K) - b_K \le \mathbb{P}(A_r \ge y) \le \mathbb{P}(rZ_r \ge y - 2K) + b_K,$$

and rZ_r has the law of rB by Proposition B.3. So $\mathbb{P}(A_r \ge y)$ and $\mathbb{P}(rB \ge y)$ differ by at most b_K plus the probability that rB lies in an interval of length $2K$. Hence the left side of the lemma is at most $b_K + \sup_x \mathbb{P}(x \le B \le x + 2K/r)$.

Step 4 (short intervals and the choice of K). We bound the mass that B gives to an interval of length δ. For $c \le 1$ the density of B is at most $2^{1-c}\bigl(x^{c-1} + (1-x)^{c-1}\bigr)/\mathrm{B}(c,c)$, and x^{c-1} has integral at most δ^c/c over such an interval. For $c \ge 1$ the density is at most d_c. The mass of an interval is also at most 1. So there is C'_J such that an interval of length δ has mass at most $C'_J\delta^{a_J}$ for all $\delta > 0$ and $c \in J$. A large K makes b_K small and the interval long, and we balance the two. Take $K = \lceil r^{\kappa_J} \rceil$. Then $b_K \le (3 + 2\log r)r^{-\kappa_J}$ and $2K/r \le 4r^{\kappa_J - 1}$, and $(r^{\kappa_J - 1})^{a_J} = r^{-\kappa_J}$ because $(1-\kappa_J)a_J = \kappa_J$. So both terms are at most a constant times $(\log r)\, r^{-\kappa_J}$. □

B.4 A local limit at the center

Proposition B.5 (local limit at the center)**.** *For each compact $J \subset (0,2)$ there is C_J such that $\bigl|NC_N - d_c\bigr| \le C_J\,(\log N)\,N^{-\kappa_J}$ for $c \in J$ and even $N \ge 4$.*

Proof. We start from the averaging identity of Lemma B.1 and replace the law of $T = T_h^+$ by its continuum version. Since $T \le r$ exactly when $A_r \ge h$, and A_r is close to rB by Lemma B.4, the

time T is close in law to h/B. The weights $\lambda_{r+1,N}$ are explicit, so the average becomes an explicit integral against the Beta(c,c) density.

Let $F(r)=\mathbb{P}(T\le r)=\mathbb{P}(A_r\ge h)$, $\Phi(r)=\mathbb{P}(rB\ge h)$ and $\Lambda_r=N\lambda_{r+1,N}$. By Lemma B.1,

$$NC_N=2\sum_{r=h}^{N-1}\big(F(r)-F(r-1)\big)\Lambda_r.$$

Step 1 (replacing F by Φ). Put $\Delta(r)=F(r)-\Phi(r)$. Then $\Delta(h-1)=0$, because $A_{h-1}<h$ and $B<1$. For $h\le r\le N-1$ we have $r\ge 2$ and $N/2\le r<N$, so Lemma B.4 gives $|\Delta(r)|\le\delta_N:=C_J(\log N)N^{-\kappa_J}$ after enlarging C_J. Summation by parts gives

$$\sum_{r=h}^{N-1}\big(\Delta(r)-\Delta(r-1)\big)\Lambda_r=\Delta(N-1)\Lambda_{N-1}-\sum_{r=h}^{N-2}\Delta(r)\big(\Lambda_{r+1}-\Lambda_r\big).$$

The ratio $\lambda_{t+1,N}/\lambda_{t,N}=t/(t+1-c)$ shows that Λ_r is monotone in r (it decreases for $c\le 1$ and increases for $c\ge 1$). So the differences $\Lambda_{r+1}-\Lambda_r$ have one sign, and their absolute values add up to $|\Lambda_{N-1}-\Lambda_h|$. Hence the right side is at most $\delta_N\big(\Lambda_{N-1}+|\Lambda_{N-1}-\Lambda_h|\big)$ in absolute value. Now $\Lambda_{N-1}=N\lambda_{N,N}=c$ and $0<\Lambda_h<2c$. For $c\le 1$ the latter holds because $\Lambda_h=\frac{Nc}{h+1}\rho(h+1,N)$ with $\rho\le 1$ and $N<2(h+1)$, and for $c>1$ because Λ_r increases to c. So $\Lambda_{N-1}+|\Lambda_{N-1}-\Lambda_h|\le 2c$, and replacing F by Φ changes NC_N by at most $4c\,\delta_N$.

Step 2 (the new sum is an expectation). After the replacement the sum is $2\sum_{r=h}^{N-1}(\Phi(r)-\Phi(r-1))\Lambda_r$. The term with $r=h$ vanishes, since $\Phi(h)=\Phi(h-1)=0$. For $h<r\le N-1$ we have $\Phi(r)-\Phi(r-1)=\mathbb{P}(h/r\le B<h/(r-1))$, and on this event $\lceil h/B\rceil=r$. These events are disjoint, and their union is $\{h/(N-1)\le B<1\}$. So the new sum is $2\,\mathbb{E}\big[\Lambda_{\lceil h/B\rceil};\,B\ge h/(N-1)\big]$.

Step 3 (evaluating the expectation). By (9) with $y=N$ and $y=r+1\ge 3$, $\Lambda_r=c\big((r+1)/N\big)^{c-1}(1+O(1/N))$ for $h\le r\le N-1$. On the event above with $r=\lceil h/B\rceil$, the number $(r+1)/N$ lies in $[\frac12,2]$ and within $2/N$ of $h/(BN)=1/(2B)$, which also lies in $[\frac12,2]$. The function x^{c-1} is 4-Lipschitz on $[\frac12,2]$. So $\Lambda_{\lceil h/B\rceil}=c(2B)^{1-c}+O(1/N)$. Also $\mathbb{P}\big(\frac12\le B<h/(N-1)\big)=O_J(1/N)$, since the density of B is bounded near $\frac12$ uniformly on J. So we may replace the event $\{B\ge h/(N-1)\}$ by $\{B\ge\frac12\}$. In the remaining integral the factor b^{1-c} cancels the factor b^{c-1} of the density. Hence NC_N is within $4c\,\delta_N$ of

$$2c\,\mathbb{E}\big[(2B)^{1-c};\,B\ge\tfrac12\big]+O_J(1/N)=\frac{2c\,2^{1-c}}{\mathrm{B}(c,c)}\int_{1/2}^1(1-b)^{c-1}\,db+O_J(1/N)=d_c+O_J(1/N).\quad\square$$

B.5 The expansion of the crossing

Proof of Proposition 3.7. We write $\log C_N-\log E_N$ as an explicit function of c with a small error, and we solve for its zero near $c=1$. Let $J=[c^{(3)},1]$. Then $\kappa_J=c^{(3)}/(1+c^{(3)})>0.42$, and d_c is bounded below on J by a positive constant. By (9) with $y=N$, $\log E_N=\log\big(\frac12\rho(1,N)\big)=-\log 2-\log\Gamma(2-c)-c\log N+O(1/N)$. By Proposition B.5, $\log C_N=\log d_c-\log N+O(N^{-0.4})$ for large N. Both hold uniformly for $c\in J$, so

$$\log C_N-\log E_N=(c-1)\log N+\Theta(c)+O(N^{-0.4})\qquad(c\in J).\tag{11}$$

Here $\Theta(1)=\log 2$ because $d_1=1$. Next, $\Theta'(c)=-\psi(2-c)-\log 4-2\psi(c)+2\psi(2c)$, and with $\psi(1)=-\gamma$ and $\psi(2)=1-\gamma$ [22, 5.4.12, 5.4.14] this gives $\Theta'(1)=\gamma+2-2\log 2$. Put $L=\log N+\Theta'(1)$. By Proposition B.2, $c_N\in J$ for $N\ge 4$ and $\varepsilon=1-c_N$ is $O(1/\log N)$. The left

side of (11) vanishes at $c = c_N$, because $C_N = E_N$ there. As Θ is smooth on J, Taylor's theorem gives $\Theta(c_N) = \log 2 - \varepsilon\,\Theta'(1) + O(\varepsilon^2)$. So (11) at $c = c_N$ reads $0 = -\varepsilon \log N + \log 2 - \varepsilon\,\Theta'(1) + O(\varepsilon^2) + O(N^{-0.4})$, that is $\varepsilon L = \log 2 + O(\varepsilon^2) + O(N^{-0.4}) = \log 2 + O(L^{-2})$. So $\varepsilon = \log 2/L + O(L^{-3})$, which is the expansion. □

With one more term in Taylor's theorem and $\varepsilon^2 = (\log 2)^2/L^2 + O(L^{-4})$, the same argument adds the term $-\Theta''(1)(\log 2)^2/(2L^3)$ to the expansion of c_N, with error $O(L^{-4})$. Here $\Theta''(1) = \pi^2/2 - 4$, since $\Theta''(c) = \psi'(2-c) - 2\psi'(c) + 4\psi'(2c)$, $\psi'(1) = \zeta(2)$ and $\psi'(2) = \zeta(2) - 1$ [22, 5.4.12, 5.15.5]. At $N = 200$ the bracket of Proposition B.2 is $[0.88749, 0.89878]$, Proposition 3.7 gives 0.89319, and with the L^{-3} term we get 0.89236. The high-precision value is $c_{200} = 0.89253$. The constants C_J and the constants in the O-terms of this appendix are not computed.

Statements and Declarations

Funding. No funding was received for this work.

Competing interests. The author has no competing interests to declare.

Ethics approval and consent. Not applicable.

Data availability. This paper uses no external data. The data computed for it (the cached data of Figure 1 and the prediction ledgers) are in the folder `kagey-problems/problem131/paperF` of `https://github.com/apemm/Kagey-Problems`, at the release `paperF-arxiv-v1`.

Code availability. All code and Lean files supporting this paper are in the folder `kagey-problems/problem131` of the same repository, at the same release. To rerun the checks, run `python -B verify_all.py --paper f` in that folder, which takes a few minutes. With the option `--full` it also runs the heavy cells (among them the formula of Proposition 2.2 at $N = 10^5$), and it then takes about twenty minutes. The scripts are in the subfolder `paperF` and need `numpy`, `scipy` and `mpmath`. Figure 1 is made by `paperD/figures/make_figures.py` and Figure 2 by `paperF/figures/make_aging_figure.py`. The file `SUPPLEMENT.md` says what each script checks and where to find the predictions that we wrote down before each computation. Section 4 describes the checks.

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